Ch 12 · Chronic Neurologic Disorders

ALZHEIMER'S DISEASE

Donepezil *(Aricept)*

Ch 12 · Alzheimer's disease

Cholinesterase inhibitor (acetylcholinesterase inhibitor) · also: Rivastigmine (oral or transdermal patch), galantamine

How it works: Blocks acetylcholinesterase so more acetylcholine stays available at brain synapses, improving memory/thinking.

INDICATIONS
- Mild, moderate, and severe Alzheimer's disease (symptom relief)

MAIN SIDE EFFECTS
- Nausea, vomiting, diarrhea, anorexia, weight loss
- Dizziness, headache
- Insomnia, vivid dreams

ADVERSE EFFECTS
- Bradycardia, syncope (vagal effect) — monitor HR, fall precautions
- GI bleeding (increased gastric acid) — report black/tarry stools
- Muscle weakness, possible seizures

CONTRAINDICATIONS & PRECAUTIONS
- Caution: asthma/COPD (chronic obstructive pulmonary disease) — bronchoconstriction
- Caution: bradycardia, sick sinus syndrome, heart block
- Caution: peptic ulcer disease, GI bleeding
- Caution: seizure disorder, urinary obstruction

INTERACTIONS
- Anticholinergics (e.g., benztropine, diphenhydramine) → cancel benefit; avoid
- NSAIDs (nonsteroidal anti-inflammatory drugs) → GI bleed risk; monitor
- Succinylcholine → prolonged neuromuscular blockade; tell anesthesia

SPECIAL WARNINGS
- Does NOT cure or stop disease — slows symptom decline
- Benefit takes weeks; reassess regularly
- Monitor HR, weight, GI tolerance

CLIENT EDUCATION
- Take once daily at bedtime, with or without food
- Do not stop suddenly — function can drop quickly
- Rise slowly; report fainting or slow pulse
- Report black stools or vomiting blood

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Cholinesterase inhibitor + anticholinergic = they cancel each other out.

Memantine *(Namenda)*

Ch 12 · Alzheimer's disease

NMDA (N-methyl-D-aspartate) receptor antagonist

How it works: Blocks NMDA receptors so excess glutamate (calcium influx) doesn't damage neurons; improves learning/memory.

INDICATIONS
- Moderate to severe Alzheimer's disease (often combined with donepezil)

MAIN SIDE EFFECTS
- Dizziness, headache
- Confusion
- Constipation

ADVERSE EFFECTS
- Hallucinations or worsening confusion — report, ensure safety

CONTRAINDICATIONS & PRECAUTIONS
- Caution: severe renal impairment — dose reduction
- Caution: conditions that raise urine pH

INTERACTIONS
- Urine alkalinizers (sodium bicarbonate) → ↓ memantine excretion, ↑ levels; monitor for toxicity

SPECIAL WARNINGS
- Not a cure; slows symptom decline
- Monitor renal function

CLIENT EDUCATION
- Can take with or without food
- Rise slowly, use fall precautions for dizziness
- Caregiver: report new confusion or hallucinations

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Donepezil = more ACh; memantine = less glutamate. Different mechanisms, often used together.

PARKINSON'S DISEASE

Carbidopa/levodopa *(Sinemet)*

Ch 12 · Parkinson's disease

Dopamine replacement (dopaminergic)

How it works: Levodopa crosses BBB (blood-brain barrier) → converted to dopamine; carbidopa blocks peripheral conversion so more reaches brain.

INDICATIONS
- Parkinson's disease

MAIN SIDE EFFECTS
- Nausea/vomiting, anorexia
- Orthostatic hypotension
- Darkened sweat and urine (harmless)

ADVERSE EFFECTS
- Dyskinesias (abnormal involuntary movements) — may need dose reduction
- Psychosis: hallucinations, nightmares, paranoia
- Cardiac dysrhythmias (palpitations) — monitor HR/rhythm
- Sudden sleep onset; impulse-control problems (gambling, spending)

CONTRAINDICATIONS & PRECAUTIONS
- Narrow-angle glaucoma
- History of melanoma or undiagnosed skin lesions
- Nonselective MAOI (monoamine oxidase inhibitor) use within 2 weeks
- Caution: cardiac disease, psychiatric disorders

INTERACTIONS
- Nonselective MAOIs → hypertensive crisis; stop MAOI 2 weeks before
- First-generation antipsychotics (haloperidol) → block dopamine, ↓ effect; avoid
- Anticholinergics → enhance effect (also ↑ anticholinergic SE)
- Pyridoxine (vitamin B6) → ↓ levodopa effect
- High-protein meals → ↓ absorption/brain uptake

SPECIAL WARNINGS
- Wearing-off / on-off phenomenon — may need dose changes or adjunct drugs
- Full benefit may take weeks to months
- Assess for orthostatic BP, dyskinesias, mental status changes

CLIENT EDUCATION
- Take with food to reduce nausea, but avoid high-protein meals
- Spread protein across the day
- Change positions slowly
- Sweat/urine may turn dark — harmless
- Don't drive until you know if sudden sleepiness occurs

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Pyridoxine (B6) speeds peripheral levodopa breakdown — avoid B6 supplements and high-protein meals.

Pramipexole *(Mirapex)*

Ch 12 · Parkinson's disease

Dopamine agonist (non-ergot) · also: Ropinirole (Requip), rotigotine patch

How it works: Directly stimulates dopamine receptors in the brain — no conversion needed.

INDICATIONS
- Parkinson's disease (alone early, or with levodopa)
- Restless legs syndrome

MAIN SIDE EFFECTS
- Nausea
- Daytime sleepiness, dizziness
- Orthostatic hypotension

ADVERSE EFFECTS
- Sudden sleep attacks (falling asleep during activities) — no driving if occurs
- Hallucinations, confusion (esp. older adults)
- Dyskinesias when combined with levodopa
- Impulse-control disorders: gambling, shopping, hypersexuality — report

CONTRAINDICATIONS & PRECAUTIONS
- Caution: renal impairment (pramipexole) — reduce dose
- Caution: older adults, psychiatric history

INTERACTIONS
- Levodopa → ↑ dyskinesias, hallucinations; levodopa dose may be lowered
- CNS (central nervous system) depressants/alcohol → ↑ sedation; avoid

SPECIAL WARNINGS
- Assess for sleep attacks and compulsive behaviors at each visit
- Monitor orthostatic BP

CLIENT EDUCATION
- Avoid driving/hazardous activity until you know how drug affects you
- Report urges to gamble or spend uncontrollably
- Change positions slowly
- Take with food to reduce nausea

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Agonists act directly on receptors — can be used before levodopa to delay its long-term problems.

Selegiline *(Eldepryl, Zelapar)*

Ch 12 · Parkinson's disease

MAO-B (monoamine oxidase type B) inhibitor · also: Rasagiline (Azilect), safinamide

How it works: Selectively blocks MAO-B, which breaks down dopamine in the brain → dopamine lasts longer.

INDICATIONS
- Parkinson's disease (adjunct to levodopa or early monotherapy)

MAIN SIDE EFFECTS
- Insomnia (selegiline metabolizes to amphetamine)
- Dizziness, orthostatic hypotension
- Nausea, dry mouth

ADVERSE EFFECTS
- Hypertensive crisis at high doses (loses MAO-B selectivity) with tyramine
- Increased levodopa adverse effects (dyskinesias, hallucinations)

CONTRAINDICATIONS & PRECAUTIONS
- Use with meperidine — contraindicated
- Caution: use with other serotonergic drugs

INTERACTIONS
- Meperidine → serotonin syndrome / possible death; never combine
- SSRIs (selective serotonin reuptake inhibitors), TCAs (tricyclic antidepressants) → serotonin syndrome; avoid
- Levodopa → ↑ effect; levodopa dose may be reduced
- Tyramine-rich foods (at high doses) → hypertensive crisis

SPECIAL WARNINGS
- Selective for MAO-B only at recommended doses
- Monitor BP; teach hypertensive crisis signs (severe headache, stiff neck)

CLIENT EDUCATION
- Take selegiline in the morning (or early) — prevents insomnia
- Do not exceed prescribed dose
- Report severe headache, chest pain, racing heart
- Tell all providers you take this — many drug interactions

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Selegiline → amphetamine metabolites → insomnia; dose early in the day.

Entacapone *(Comtan (with carbidopa/levodopa = Stalevo))*

Ch 12 · Parkinson's disease

COMT (catechol-O-methyltransferase) inhibitor · also: Tolcapone, opicapone

How it works: Blocks COMT breakdown of levodopa in the periphery → more levodopa reaches brain, longer 'on' time.

INDICATIONS
- Parkinson's disease — adjunct to levodopa for wearing-off

MAIN SIDE EFFECTS
- Brownish-orange urine discoloration (harmless)
- Nausea, diarrhea
- Orthostatic hypotension

ADVERSE EFFECTS
- Dyskinesias, hallucinations (↑ levodopa effects) — levodopa dose may be lowered
- Sudden sleep onset

CONTRAINDICATIONS & PRECAUTIONS
- Caution: liver impairment

INTERACTIONS
- Nonselective MAOIs → may ↑ catecholamines; avoid combination
- Levodopa → ↑ effects and adverse effects (intended)

SPECIAL WARNINGS
- No effect alone — must be given WITH levodopa
- Tolcapone carries risk of liver failure (monitor LFTs, liver function tests)

CLIENT EDUCATION
- Take each dose with your carbidopa/levodopa
- Urine may turn orange-brown — harmless
- Change positions slowly

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: COMT inhibitor only works as a levodopa 'booster' — useless on its own.

Amantadine *(Gocovri (ER), Symmetrel)*

Ch 12 · Parkinson's disease

Dopamine-releasing agent (antiviral)

How it works: Promotes dopamine release and may block reuptake; also NMDA antagonist that reduces dyskinesias.

INDICATIONS
- Parkinson's disease (early, and levodopa-induced dyskinesias)
- Influenza A (antiviral origin; resistance common)

MAIN SIDE EFFECTS
- Dizziness, orthostatic hypotension
- Anticholinergic-like: dry mouth, constipation, urinary retention
- Insomnia, nervousness

ADVERSE EFFECTS
- Livedo reticularis (mottled purplish skin, usually legs) — reversible on stopping
- Confusion, hallucinations (esp. older adults)
- Peripheral edema

CONTRAINDICATIONS & PRECAUTIONS
- Caution: renal impairment — dose reduction
- Caution: seizure history, HF (heart failure)

INTERACTIONS
- Anticholinergics → ↑ anticholinergic effects; monitor
- CNS stimulants/alcohol → ↑ CNS effects; avoid

SPECIAL WARNINGS
- Rapid onset (days) but benefit may fade within months
- Taper — abrupt stop can worsen Parkinson's symptoms

CLIENT EDUCATION
- Report mottled skin, swelling, confusion
- Avoid alcohol
- Don't stop abruptly

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Livedo reticularis = classic amantadine finding.

Benztropine *(Cogentin)*

Ch 12 · Parkinson's disease

Centrally acting anticholinergic · also: Trihexyphenidyl

How it works: Blocks muscarinic acetylcholine receptors in the brain → rebalances dopamine/ACh, reducing tremor and rigidity.

INDICATIONS
- Parkinson's disease (mainly tremor)
- Drug-induced EPS (extrapyramidal symptoms) from antipsychotics

MAIN SIDE EFFECTS
- Dry mouth, blurred vision, photophobia
- Constipation, urinary retention
- Tachycardia
- Decreased sweating → overheating

ADVERSE EFFECTS
- Confusion, hallucinations, delirium (esp. older adults)
- Heat stroke/hyperthermia
- Anticholinergic toxicity — report

CONTRAINDICATIONS & PRECAUTIONS
- Narrow-angle glaucoma
- BPH (benign prostatic hyperplasia)/urinary retention
- GI obstruction, paralytic ileus
- Myasthenia gravis
- Caution: older adults (Beers list)

INTERACTIONS
- Antihistamines, TCAs, phenothiazines → additive anticholinergic effects; avoid
- Cholinesterase inhibitors (donepezil) → opposing effects

SPECIAL WARNINGS
- Monitor I&O (intake and output), bowel function, mental status

CLIENT EDUCATION
- Sugarless gum/sips of water for dry mouth
- Sunglasses for light sensitivity
- Void before each dose; report trouble urinating
- Increase fiber/fluids for constipation
- Avoid hot environments and strenuous exercise in heat

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Anticholinergic: 'can't see, can't pee, can't spit, can't poop' — and can't sweat.

MULTIPLE SCLEROSIS

Interferon beta-1a / beta-1b *(Avonex, Rebif (1a); Betaseron, Extavia (1b))*

Ch 12 · Multiple sclerosis

Immunomodulator (disease-modifying therapy) · also: Peginterferon beta-1a (Plegridy)

How it works: Modulates the immune response to reduce inflammation and attacks on myelin; decreases relapses.

INDICATIONS
- Relapsing forms of MS (multiple sclerosis)

MAIN SIDE EFFECTS
- Flu-like symptoms: fever, chills, myalgia, fatigue
- Injection-site reactions (redness, pain)
- Headache

ADVERSE EFFECTS
- Depression, suicidal ideation — screen and report
- Hepatotoxicity — monitor LFTs (liver function tests)
- Myelosuppression (↓ WBC, platelets) — monitor CBC (complete blood count)
- Injection-site necrosis

CONTRAINDICATIONS & PRECAUTIONS
- Caution: depression/suicidal history
- Caution: liver disease, seizure disorders
- Caution: pregnancy

INTERACTIONS
- Other hepatotoxic or myelosuppressive drugs → additive toxicity; monitor

SPECIAL WARNINGS
- Monitor LFTs and CBC at baseline and periodically
- Avonex = IM (intramuscular) weekly; Rebif SC (subcutaneous) 3x/week; Betaseron SC every other day

CLIENT EDUCATION
- Premedicate with acetaminophen or ibuprofen; inject at bedtime to sleep through flu-like symptoms
- Rotate injection sites
- Report sadness, hopelessness, or yellow skin/eyes
- Report signs of infection or unusual bleeding

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Interferon = flu-like symptoms + depression; glatiramer has neither.

Glatiramer acetate *(Copaxone)*

Ch 12 · Multiple sclerosis

Immunomodulator (disease-modifying therapy)

How it works: Synthetic myelin-like protein acts as a decoy, shifting T cells away from attacking myelin.

INDICATIONS
- Relapsing forms of MS

MAIN SIDE EFFECTS
- Injection-site reactions (pain, redness, itching)
- Lipoatrophy at injection sites

ADVERSE EFFECTS
- Post-injection reaction: flushing, chest pain, palpitations, dyspnea, anxiety — transient, resolves without treatment

CONTRAINDICATIONS & PRECAUTIONS
- Hypersensitivity to glatiramer or mannitol

SPECIAL WARNINGS
- Given SC daily (20 mg) or 3x/week (40 mg)
- No flu-like syndrome, depression, or lab monitoring like interferons

CLIENT EDUCATION
- Rotate injection sites to prevent skin dents (lipoatrophy)
- Flushing/chest tightness after a dose usually passes in minutes
- Seek help if chest pain persists or breathing worsens

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Post-injection reaction is frightening but brief — reassure, observe.

Natalizumab *(Tysabri)*

Ch 12 · Multiple sclerosis

Monoclonal antibody (integrin inhibitor, immunosuppressant)

How it works: Blocks immune cells from crossing into the CNS, so fewer attack myelin.

INDICATIONS
- Relapsing forms of MS
- Crohn's disease

MAIN SIDE EFFECTS
- Headache, fatigue, joint pain
- Infusion reactions

ADVERSE EFFECTS
- PML (progressive multifocal leukoencephalopathy) — fatal brain infection by JC virus
- Hypersensitivity/anaphylaxis during infusion — stop infusion
- Hepatotoxicity — monitor LFTs
- Serious infections (herpes, others)

CONTRAINDICATIONS & PRECAUTIONS
- History of PML
- Immunocompromised clients
- Caution: with other immunosuppressants — avoid

INTERACTIONS
- Immunosuppressants (corticosteroids, TNF blockers) → ↑ PML/infection risk; avoid

SPECIAL WARNINGS
- **BLACK BOX:** PML — restricted program (TOUCH); check JC virus antibodies
- IV infusion every 4 weeks; observe for reactions during and 1 hr after
- Assess neuro status before each dose

CLIENT EDUCATION
- Report new weakness, clumsiness, vision changes, confusion, or personality change immediately
- Report signs of infection, jaundice, dark urine
- Carry medication guide; keep all infusion appointments

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Natalizumab → PML: new neuro symptoms that progress over days = hold drug and report.

Fingolimod *(Gilenya)*

Ch 12 · Multiple sclerosis

S1P (sphingosine 1-phosphate) receptor modulator (oral immunomodulator) · also: Siponimod, ozanimod

How it works: Traps lymphocytes in lymph nodes so fewer reach the CNS to damage myelin.

INDICATIONS
- Relapsing forms of MS (oral)

MAIN SIDE EFFECTS
- Headache
- Diarrhea
- Back pain, cough

ADVERSE EFFECTS
- First-dose bradycardia / AV (atrioventricular) block — monitor HR and BP 6 hr after first dose
- Infections (herpes, others); rare PML
- Macular edema — eye exam baseline and 3–4 months
- Hepatotoxicity; decreased pulmonary function

CONTRAINDICATIONS & PRECAUTIONS
- Recent MI (myocardial infarction), stroke, HF, or unstable angina
- 2nd/3rd degree AV block or sick sinus without pacemaker
- Class Ia or III antidysrhythmics

INTERACTIONS
- Beta blockers, calcium channel blockers → additive bradycardia
- Live vaccines → avoid during and 2 months after
- Immunosuppressants → ↑ infection risk

SPECIAL WARNINGS
- ECG before first dose; 6-hr observation after first dose
- Check CBC, LFTs, varicella immunity before starting
- Teratogenic — effective contraception during and 2 months after stopping

CLIENT EDUCATION
- Report slow pulse, dizziness, or fainting
- Report vision changes
- Report fever or signs of infection
- Use reliable birth control

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Fingolimod first dose = heart monitor watch (bradycardia).

Mitoxantrone

Ch 12 · Multiple sclerosis

Antineoplastic immunosuppressant

How it works: Antineoplastic that suppresses T cells, B cells, and macrophages that attack myelin.

INDICATIONS
- Secondary progressive or worsening relapsing MS
- Certain leukemias and prostate cancer

MAIN SIDE EFFECTS
- Nausea, alopecia
- Blue-green urine (24 hr); bluish sclera
- Menstrual irregularities

ADVERSE EFFECTS
- Cardiotoxicity/HF — check LVEF (left ventricular ejection fraction) before each dose
- Myelosuppression — monitor CBC; infection precautions
- Secondary AML (acute myeloid leukemia)
- Extravasation → tissue damage; IV only

CONTRAINDICATIONS & PRECAUTIONS
- LVEF < 50% or cumulative lifetime dose reached
- Pregnancy (teratogen)
- Caution: liver impairment

INTERACTIONS
- Other cardiotoxic or myelosuppressive drugs → additive toxicity

SPECIAL WARNINGS
- **BLACK BOX:** cardiotoxicity — lifetime cumulative dose limit (140 mg/m^2)
- **BLACK BOX:** secondary AML
- IV infusion every 3 months; check pregnancy test before each dose

CLIENT EDUCATION
- Urine may turn blue-green for a day — harmless
- Report shortness of breath, swelling, rapid heartbeat
- Avoid crowds/sick contacts; report fever
- Use effective contraception

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Mitoxantrone: lifetime dose cap because of the heart.

High-dose corticosteroids for acute MS relapse *(Solu-Medrol (methylprednisolone))*

Ch 12 · Multiple sclerosis

Glucocorticoid · also: Oral prednisone (taper)

How it works: Reduces inflammation and swelling in the CNS, shortening the relapse.

INDICATIONS
- Acute MS exacerbation (relapse) — IV methylprednisolone ~3–5 days, ± oral taper

MAIN SIDE EFFECTS
- Insomnia, mood swings, euphoria
- Increased appetite, fluid retention
- Metallic taste

ADVERSE EFFECTS
- Hyperglycemia — monitor glucose
- Hypokalemia — monitor potassium
- Infection risk (immunosuppression)
- GI bleeding/peptic ulcer

CONTRAINDICATIONS & PRECAUTIONS
- Systemic fungal infection
- Caution: diabetes, HF, peptic ulcer, active infection

INTERACTIONS
- NSAIDs → ↑ GI bleed risk
- Potassium-wasting diuretics → hypokalemia
- Antidiabetics → ↓ effect; may need dose increase

SPECIAL WARNINGS
- Shortens relapse but does NOT change long-term disease course
- Monitor glucose, BP, electrolytes, mood

CLIENT EDUCATION
- Take oral doses with food
- Report black stools, fever, or high blood sugars
- Don't stop an oral taper abruptly

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Disease-modifying drugs prevent relapses; steroids treat the relapse itself.

SEIZURE DISORDERS

Phenytoin *(Dilantin)*

Ch 12 · Seizure disorders

Antiepileptic (hydantoin) · also: Fosphenytoin (IV prodrug)

How it works: Slows sodium influx into neurons, stabilizing membranes and stopping seizure spread.

INDICATIONS
- Tonic-clonic and focal (partial) seizures
- Status epilepticus (after benzodiazepine)

MAIN SIDE EFFECTS
- Gingival hyperplasia
- Hirsutism, coarse facial features
- Morbilliform rash

ADVERSE EFFECTS
- CNS toxicity: nystagmus (first sign), ataxia, diplopia, sedation, confusion
- IV: dysrhythmias, hypotension — cardiac monitoring
- SJS/TEN (Stevens-Johnson syndrome/toxic epidermal necrolysis) — stop at rash
- Teratogenic (fetal hydantoin syndrome); bone loss/vitamin D deficiency

CONTRAINDICATIONS & PRECAUTIONS
- Sinus bradycardia, SA (sinoatrial) block, 2nd/3rd degree AV block, Adams-Stokes
- Caution: pregnancy, liver disease, HLA-B*1502 carriers

INTERACTIONS
- Oral contraceptives, warfarin, glucocorticoids → ↓ effect (enzyme induction); use backup birth control
- Alcohol, other CNS depressants → ↑ CNS depression
- Enteral tube feedings → ↓ absorption; hold feeds 1–2 hr before/after
- Valproate, cimetidine, isoniazid → ↑ phenytoin levels

SPECIAL WARNINGS
- Therapeutic level 10–20 mcg/mL (narrow; small dose change → big level change)
- BLACK BOX (IV): cardiovascular risk with rapid infusion — max 50 mg/min adults
- IV: mix in normal saline ONLY (precipitates in dextrose); flush line with saline before/after
- Purple glove syndrome (IV extravasation) — use large vein, check site
- FDA class warning: suicidal thoughts — monitor mood

CLIENT EDUCATION
- Brush, floss, and see dentist regularly; use soft toothbrush
- Don't switch brands without provider approval
- Use non-hormonal backup contraception
- Get regular blood levels; report unsteady walking or double vision
- Don't stop suddenly

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Phenytoin IV: Saline only, Slow (≤50 mg/min), watch for purple glove.

Fosphenytoin *(Cerebyx)*

Ch 12 · Seizure disorders

Antiepileptic (hydantoin prodrug)

How it works: IV/IM prodrug converted to phenytoin in the body; same sodium-channel action.

INDICATIONS
- Status epilepticus (after benzodiazepine)
- Short-term replacement when oral phenytoin can't be given

MAIN SIDE EFFECTS
- Pruritus/paresthesias (groin) during infusion
- Dizziness, sedation
- Nystagmus, ataxia

ADVERSE EFFECTS
- Hypotension and dysrhythmias with rapid IV — cardiac/BP monitoring
- Same toxicity as phenytoin

CONTRAINDICATIONS & PRECAUTIONS
- Sinus bradycardia, SA block, 2nd/3rd degree AV block, Adams-Stokes

INTERACTIONS
- Same as phenytoin (enzyme inducer: ↓ oral contraceptives, warfarin)

SPECIAL WARNINGS
- Dosed in PE (phenytoin sodium equivalents) — read order carefully
- **BLACK BOX:** cardiovascular risk with rapid infusion — max 150 mg PE/min
- Can be given IV or IM; compatible with saline OR dextrose
- Less tissue injury/purple glove than IV phenytoin
- Monitor phenytoin level (10–20 mcg/mL)

CLIENT EDUCATION
- Tell nurse about itching, tingling, dizziness during infusion

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Fosphenytoin vs phenytoin: faster rate, IM option, dextrose OK, gentler on veins.

Carbamazepine *(Tegretol)*

Ch 12 · Seizure disorders

Antiepileptic · also: Oxcarbazepine (Trileptal)

How it works: Slows sodium influx into neurons, suppressing repetitive firing and seizure spread.

INDICATIONS
- Focal (partial) and tonic-clonic seizures (not absence)
- Trigeminal neuralgia
- Bipolar disorder

MAIN SIDE EFFECTS
- Dizziness, drowsiness, ataxia, diplopia, blurred vision
- Nausea, vomiting

ADVERSE EFFECTS
- Aplastic anemia, agranulocytosis, thrombocytopenia — monitor CBC
- SJS/TEN — stop drug at first rash
- Hyponatremia (SIADH-like water retention) — monitor sodium
- Teratogenic (neural tube defects)

CONTRAINDICATIONS & PRECAUTIONS
- Bone marrow suppression
- MAOI use within 14 days
- Caution: pregnancy, liver disease, Asian ancestry with HLA-B*1502

INTERACTIONS
- Grapefruit juice → ↑ carbamazepine levels; avoid
- Oral contraceptives, warfarin → ↓ effect (enzyme induction)
- Induces its own metabolism (autoinduction) — dose adjusted over weeks
- Phenytoin, phenobarbital → ↓ carbamazepine levels

SPECIAL WARNINGS
- Therapeutic level 4–12 mcg/mL
- **BLACK BOX:** SJS/TEN — test for HLA-B*1502 before starting in clients of Asian ancestry
- **BLACK BOX:** aplastic anemia/agranulocytosis — baseline and periodic CBC
- FDA class warning: suicidal thoughts

CLIENT EDUCATION
- Take with food
- Report sore throat, fever, bruising, bleeding, or any rash
- Avoid grapefruit juice
- Use non-hormonal birth control
- Don't drive until you know how it affects you

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Carbamazepine: check HLA-B*1502 (skin) and CBC (bone marrow) — both black box.

Valproic acid *(Depakene, Depakote (divalproex))*

Ch 12 · Seizure disorders

Antiepileptic (broad-spectrum)

How it works: Blocks sodium and calcium channels and boosts GABA (gamma-aminobutyric acid) → suppresses neuron firing.

INDICATIONS
- All seizure types incl. absence, myoclonic, tonic-clonic
- Bipolar disorder (mania)
- Migraine prevention

MAIN SIDE EFFECTS
- Nausea, vomiting, indigestion
- Weight gain
- Tremor, hair thinning

ADVERSE EFFECTS
- Hepatotoxicity — monitor LFTs; highest risk under age 2
- Pancreatitis — report severe abdominal pain, N/V
- Thrombocytopenia — monitor platelets/bleeding
- Hyperammonemia — lethargy, confusion; check ammonia

CONTRAINDICATIONS & PRECAUTIONS
- Liver disease
- Urea cycle disorders
- Pregnancy (especially for migraine prevention)

INTERACTIONS
- Phenytoin, phenobarbital → valproate ↑ their levels; monitor levels
- Lamotrigine → valproate ↑ lamotrigine levels (SJS risk); lower lamotrigine dose
- Carbapenem antibiotics → ↓ valproate levels (seizure risk)
- Alcohol/CNS depressants → ↑ sedation

SPECIAL WARNINGS
- Therapeutic level 50–100 mcg/mL
- **BLACK BOX:** hepatotoxicity (life-threatening)
- **BLACK BOX:** pancreatitis
- **BLACK BOX:** teratogen — neural tube defects, lower IQ; folic acid, effective contraception
- FDA class warning: suicidal thoughts

CLIENT EDUCATION
- Report yellow skin/eyes, dark urine, abdominal pain, vomiting
- Report unusual bruising or bleeding
- Take with food; swallow delayed-release tabs whole
- Use reliable birth control; tell provider if planning pregnancy

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Valproate black box = liver, pancreas, pregnancy.

Lamotrigine *(Lamictal)*

Ch 12 · Seizure disorders

Antiepileptic

How it works: Blocks sodium channels and reduces glutamate release, stabilizing neurons.

INDICATIONS
- Focal, tonic-clonic, absence, and Lennox-Gastaut seizures
- Bipolar disorder maintenance

MAIN SIDE EFFECTS
- Dizziness, headache
- Diplopia, blurred vision
- Nausea, vomiting

ADVERSE EFFECTS
- Serious rash/SJS/TEN — stop drug at first rash and notify provider
- Aseptic meningitis (fever, stiff neck, headache)

CONTRAINDICATIONS & PRECAUTIONS
- Caution: use with valproate (dose reduction required)
- Caution: pregnancy

INTERACTIONS
- Valproic acid → ↑ lamotrigine levels (↑ rash risk); reduce lamotrigine dose
- Carbamazepine, phenytoin, phenobarbital → ↓ lamotrigine levels
- Oral contraceptives (estrogen) → ↓ lamotrigine levels; dose may need adjustment

SPECIAL WARNINGS
- **BLACK BOX:** serious rash including SJS — highest risk early, in children, with valproate, or fast dose increases
- Must titrate slowly over weeks
- If restarting after missed doses, may need to re-titrate
- FDA class warning: suicidal thoughts

CLIENT EDUCATION
- Report ANY rash, blisters, mouth sores, or fever right away
- Follow the slow dose schedule exactly
- Don't drive until you know how vision/dizziness are affected
- Tell provider before starting or stopping birth control pills

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Lamotrigine: slow titration + rash = stop. Valproate doubles its level.

Levetiracetam *(Keppra)*

Ch 12 · Seizure disorders

Antiepileptic (newer generation)

How it works: Binds synaptic vesicle protein SV2A, reducing abnormal neurotransmitter release; exact mechanism unclear.

INDICATIONS
- Focal, myoclonic, and tonic-clonic seizures
- Status epilepticus (IV, second-line)

MAIN SIDE EFFECTS
- Drowsiness, dizziness
- Fatigue, weakness (asthenia)
- Headache, nasopharyngitis

ADVERSE EFFECTS
- Behavioral changes: agitation, irritability, aggression, depression, psychosis — report
- Suicidal thoughts
- Coordination problems

CONTRAINDICATIONS & PRECAUTIONS
- Caution: renal impairment — dose reduction
- Caution: psychiatric history

INTERACTIONS
- CNS depressants/alcohol → ↑ sedation
- Few drug-drug interactions (not metabolized by liver CYP enzymes)

SPECIAL WARNINGS
- No routine serum level monitoring
- FDA class warning: suicidal thoughts — monitor mood/behavior

CLIENT EDUCATION
- Report mood changes, anger outbursts, or depression
- Avoid alcohol
- Don't drive until you know how it affects you
- Don't stop suddenly

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Keppra 'rage' — watch for irritability and behavior changes.

Phenobarbital

Ch 12 · Seizure disorders

Antiepileptic (barbiturate), Schedule IV · also: Primidone (converted to phenobarbital)

How it works: Enhances GABA inhibition, depressing neuron firing throughout the CNS.

INDICATIONS
- Tonic-clonic and focal seizures
- Status epilepticus (refractory)
- Neonatal seizures

MAIN SIDE EFFECTS
- Drowsiness, sedation
- Cognitive slowing
- Paradoxical excitement/hyperactivity in children and older adults

ADVERSE EFFECTS
- Respiratory depression (esp. with alcohol/CNS depressants)
- Toxicity: nystagmus, ataxia, coma — supportive care
- Physical dependence; abrupt stop → withdrawal seizures
- Teratogenic; SJS rarely

CONTRAINDICATIONS & PRECAUTIONS
- Severe respiratory disease
- Porphyria
- Caution: history of substance use, depression, liver disease

INTERACTIONS
- Alcohol, opioids, benzodiazepines → additive CNS/respiratory depression; avoid
- Oral contraceptives, warfarin → ↓ effect (enzyme induction)
- Valproic acid → ↑ phenobarbital levels

SPECIAL WARNINGS
- Therapeutic level 15–40 mcg/mL
- Monitor respiratory rate, sedation level
- Taper slowly to discontinue
- FDA class warning: suicidal thoughts

CLIENT EDUCATION
- Avoid alcohol and other sedatives
- Don't drive until sedation is known
- Use non-hormonal birth control
- Never stop abruptly

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Barbiturate = no reversal agent (unlike benzos/flumazenil) — supportive airway care.

Gabapentin *(Neurontin)*

Ch 12 · Seizure disorders

Antiepileptic (newer generation, GABA analog) · also: Pregabalin (Lyrica)

How it works: Binds calcium channels in the CNS, reducing excitatory neurotransmitter release.

INDICATIONS
- Adjunct for focal (partial) seizures
- Postherpetic neuralgia / neuropathic pain

MAIN SIDE EFFECTS
- Drowsiness, dizziness
- Ataxia, fatigue
- Peripheral edema, weight gain

ADVERSE EFFECTS
- Respiratory depression with opioids/CNS depressants or in lung disease
- Suicidal thoughts
- Misuse potential

CONTRAINDICATIONS & PRECAUTIONS
- Caution: renal impairment — dose reduction
- Caution: older adults, respiratory disease

INTERACTIONS
- Antacids (aluminum/magnesium) → ↓ absorption; take gabapentin 2 hr after
- Opioids, alcohol, CNS depressants → ↑ sedation/respiratory depression

SPECIAL WARNINGS
- No routine serum level monitoring
- Monitor respiratory status when given with opioids
- FDA class warning: suicidal thoughts

CLIENT EDUCATION
- Take at least 2 hours after antacids
- Avoid alcohol; don't drive until sedation is known
- Don't stop suddenly — taper

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Gabapentin + antacid = space 2 hours.

Topiramate *(Topamax)*

Ch 12 · Seizure disorders

Antiepileptic (newer generation)

How it works: Blocks sodium channels, enhances GABA, and blocks glutamate receptors.

INDICATIONS
- Focal and tonic-clonic seizures
- Migraine prevention

MAIN SIDE EFFECTS
- Drowsiness, dizziness
- Cognitive slowing, word-finding difficulty
- Paresthesias (tingling)
- Weight loss, anorexia

ADVERSE EFFECTS
- Kidney stones — encourage fluids
- Metabolic acidosis — monitor bicarbonate
- Acute angle-closure glaucoma/myopia — report eye pain or blurred vision
- Decreased sweating → hyperthermia (esp. children)

CONTRAINDICATIONS & PRECAUTIONS
- Caution: history of kidney stones
- Caution: pregnancy (cleft lip/palate risk)

INTERACTIONS
- Oral contraceptives → ↓ effect; use backup method
- CNS depressants/alcohol → ↑ sedation
- Carbonic anhydrase inhibitors (acetazolamide) → ↑ stone/acidosis risk

SPECIAL WARNINGS
- FDA class warning: suicidal thoughts
- Monitor serum bicarbonate

CLIENT EDUCATION
- Drink 2–3 L fluid daily to prevent kidney stones
- Report eye pain or sudden vision change immediately
- Avoid overheating; stay hydrated
- Use additional/non-hormonal birth control

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Topiramate: stones, acidosis, eye pain, word-finding trouble ('Dopamax').

Ethosuximide *(Zarontin)*

Ch 12 · Seizure disorders

Antiepileptic (succinimide)

How it works: Blocks T-type calcium channels in the thalamus, suppressing absence seizure activity.

INDICATIONS
- Absence seizures ONLY (drug of choice)

MAIN SIDE EFFECTS
- Nausea, vomiting, abdominal pain
- Drowsiness, dizziness
- Headache, hiccups

ADVERSE EFFECTS
- Blood dyscrasias (rare) — monitor CBC
- SJS/serious rash
- SLE (systemic lupus erythematosus)-like syndrome
- Behavior changes, suicidal thoughts

CONTRAINDICATIONS & PRECAUTIONS
- Caution: liver or renal disease

INTERACTIONS
- Valproic acid → may alter ethosuximide levels
- CNS depressants → ↑ sedation

SPECIAL WARNINGS
- Monitor CBC periodically
- FDA class warning: suicidal thoughts

CLIENT EDUCATION
- Take with food to reduce GI upset
- Report sore throat, fever, bruising, rash, joint pain
- Don't stop suddenly

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Ethosuximide = absence seizures only; won't control tonic-clonic.

Antiepileptics — general client education

Ch 12 · Seizure disorders

Antiepileptic drugs (AEDs), all

How it works: AEDs suppress abnormal neuron firing; they control seizures but do not cure epilepsy.

INDICATIONS
- Long-term seizure prevention

MAIN SIDE EFFECTS
- CNS depression: drowsiness, dizziness, ataxia (most AEDs)

ADVERSE EFFECTS
- Suicidal thoughts/behavior (FDA warning for all AEDs) — monitor mood
- Breakthrough seizures if doses missed or stopped

CONTRAINDICATIONS & PRECAUTIONS
- Caution: pregnancy — many AEDs are teratogenic (valproate highest risk)

INTERACTIONS
- Enzyme inducers (phenytoin, carbamazepine, phenobarbital, topiramate) → ↓ oral contraceptive effect; use backup
- Alcohol, other CNS depressants → ↑ sedation

SPECIAL WARNINGS
- Never stop abruptly — can trigger status epilepticus; taper under provider
- Monitor drug levels where applicable (phenytoin, carbamazepine, valproate, phenobarbital)
- Seizure precautions: padded rails, suction and O2 at bedside

CLIENT EDUCATION
- Take at the same time every day; don't skip doses
- Keep a seizure diary (date, time, triggers, duration)
- Wear a medical alert ID
- Plan pregnancy with provider; take folic acid supplement
- Discuss birth control — some AEDs make pills less effective
- Follow state laws on driving; avoid alcohol

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: #1 AED teaching point: never stop suddenly.

Status epilepticus — IV lorazepam (first line) *(Ativan)*

Ch 12 · Seizure disorders

Benzodiazepine · also: Diazepam (Valium; IV or rectal), midazolam (IM/intranasal)

How it works: Enhances GABA inhibition, rapidly terminating seizure activity.

INDICATIONS
- Status epilepticus: seizure > 5 minutes or repeated without recovery between

MAIN SIDE EFFECTS
- Sedation, drowsiness
- Hypotension

ADVERSE EFFECTS
- Respiratory depression — have airway equipment, O2, suction ready
- Hypotension — monitor BP

CONTRAINDICATIONS & PRECAUTIONS
- Caution: respiratory compromise, other CNS depressants

INTERACTIONS
- Opioids, alcohol, barbiturates → additive respiratory depression

SPECIAL WARNINGS
- Step 1: airway, O2, side-lying, IV access, check glucose
- Step 2: IV lorazepam (or diazepam) to stop seizure
- Step 3: long-acting AED to prevent recurrence — fosphenytoin/phenytoin, valproate, or levetiracetam
- Status epilepticus is a medical emergency — risk of hypoxia and brain injury

CLIENT EDUCATION
- Family: time seizures; call 911 if > 5 minutes
- Rectal diazepam (Diastat) may be prescribed for home use

ANTIDOTE
- Flumazenil reverses benzodiazepines — use with great caution; can trigger seizures

Remember: Benzo stops the seizure now; phenytoin/fosphenytoin keeps it from coming back.

MYASTHENIA GRAVIS

Pyridostigmine *(Mestinon)*

Ch 12 · Myasthenia gravis

Reversible cholinesterase inhibitor (anticholinesterase) · also: Neostigmine

How it works: Blocks acetylcholinesterase so more ACh (acetylcholine) is available at the neuromuscular junction → stronger muscle contraction.

INDICATIONS
- Myasthenia gravis (symptom control)
- Neostigmine: reversal of nondepolarizing neuromuscular blockers, urinary retention

MAIN SIDE EFFECTS
- Increased GI motility: cramps, diarrhea, nausea
- Increased salivation, sweating
- Urinary urgency
- Miosis (pupil constriction)

ADVERSE EFFECTS
- Cholinergic crisis (too much drug) — muscle weakness, respiratory depression; give atropine, ventilate
- Bradycardia, hypotension
- Bronchospasm

CONTRAINDICATIONS & PRECAUTIONS
- Mechanical GI or urinary obstruction
- Caution: asthma, bradycardia, peptic ulcer disease, hyperthyroidism

INTERACTIONS
- Atropine/anticholinergics → ↓ effect (used as antidote)
- Succinylcholine → prolonged neuromuscular blockade

SPECIAL WARNINGS
- Dose timing is critical — give on time
- Monitor respiratory status, swallowing, muscle strength
- Have atropine and airway equipment available

CLIENT EDUCATION
- Take on exact schedule; set alarms
- Take about 30 min before meals to help chewing/swallowing
- Report trouble breathing or swallowing immediately
- Wear a medical alert ID
- Keep a diary of strength vs dose timing

ANTIDOTE
- Atropine (for cholinergic crisis)

Remember: Pyridostigmine overdose looks like SLUDGE + weakness — atropine is the antidote.

Myasthenic crisis vs cholinergic crisis

Ch 12 · Myasthenia gravis

Cholinesterase inhibitor dosing problems

How it works: Myasthenic crisis = too LITTLE ACh effect (undermedicated); cholinergic crisis = too MUCH (overmedicated).

INDICATIONS
- Distinguishing cause of acute weakness/respiratory distress in myasthenia gravis

ADVERSE EFFECTS
- Both cause severe muscle weakness and respiratory failure — airway first
- Myasthenic: weakness, ptosis, dysphagia, tachycardia, no excess secretions
- Cholinergic: weakness + SLUDGE (salivation, lacrimation, urination, diarrhea, GI cramps, emesis), bradycardia, miosis, sweating

SPECIAL WARNINGS
- Myasthenic crisis → INCREASE cholinesterase inhibitor (also plasmapheresis/IVIG)
- Cholinergic crisis → STOP cholinesterase inhibitor, give atropine, support ventilation
- Edrophonium (Tensilon) test historically: improves = myasthenic, worsens = cholinergic
- Infection, stress, surgery, missed doses can trigger myasthenic crisis

CLIENT EDUCATION
- Report increasing weakness, trouble swallowing, or shortness of breath immediately
- Report excess drooling, sweating, cramping, diarrhea (possible overdose)

ANTIDOTE
- Atropine (cholinergic crisis only)

Remember: Myasthenic = needs MORE drug; cholinergic = got TOO MUCH drug (wet, slow pulse, small pupils).

Ch 13 · Eye & Ear Disorders

GLAUCOMA

Timolol (ophthalmic) *(Timoptic)*

Ch 13 · Glaucoma

Beta-adrenergic blocker (nonselective), ophthalmic · also: Betaxolol, levobunolol, carteolol

How it works: Blocks beta receptors in the ciliary body → decreases aqueous humor production → lowers IOP (intraocular pressure).

INDICATIONS
- Open-angle glaucoma
- Ocular hypertension

MAIN SIDE EFFECTS
- Transient stinging, burning, blurred vision
- Eye irritation, dryness

ADVERSE EFFECTS
- Systemic absorption → bradycardia, hypotension — check HR/BP
- Bronchospasm (nonselective beta blockade)

CONTRAINDICATIONS & PRECAUTIONS
- Asthma, severe COPD (chronic obstructive pulmonary disease)
- Sinus bradycardia, 2nd/3rd-degree heart block, cardiogenic shock, overt heart failure
- Caution: diabetes — may mask hypoglycemia signs (tachycardia)

INTERACTIONS
- Oral beta blockers → additive beta blockade
- Verapamil, diltiazem, digoxin → additive bradycardia

SPECIAL WARNINGS
- Systemic effects are real — teach punctal occlusion
- Monitor HR, BP, and breathing in at-risk clients

CLIENT EDUCATION
- After drop, press inner corner of eye (tear duct) 1–2 min
- Report slow pulse, dizziness, wheezing, shortness of breath
- Wait at least 5 min before another eye medication

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Eye drops reach the blood — timolol can cause bradycardia and bronchospasm, so punctal occlusion matters.

Latanoprost *(Xalatan)*

Ch 13 · Glaucoma

Prostaglandin analog, ophthalmic · also: Bimatoprost, travoprost, tafluprost

How it works: Increases outflow of aqueous humor (uveoscleral) → lowers IOP (intraocular pressure).

INDICATIONS
- Open-angle glaucoma (first-line)
- Ocular hypertension

MAIN SIDE EFFECTS
- Brown pigmentation of iris (permanent), especially mixed-color eyes
- Eyelash growth — longer, thicker, darker
- Darkening of eyelid/periorbital skin
- Eye redness (conjunctival hyperemia), stinging, blurred vision

ADVERSE EFFECTS
- Macular edema, uveitis (iritis) — rare

CONTRAINDICATIONS & PRECAUTIONS
- Caution: active eye inflammation (uveitis)
- Caution: risk factors for macular edema

INTERACTIONS
- Other prostaglandin eye drops → may paradoxically raise IOP; don't combine

SPECIAL WARNINGS
- Once daily, usually in the evening — more is not better
- Iris color change may be permanent — especially notable if one eye treated

CLIENT EDUCATION
- Remove contact lenses; wait 15 min before reinserting
- Iris color may permanently change; lashes may grow
- Store unopened bottles in refrigerator
- Wait at least 5 min between different eye drops

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Latanoprost: brown eyes and long lashes (bimatoprost is even sold as Latisse for lashes).

Brimonidine *(Alphagan P)*

Ch 13 · Glaucoma

Alpha-2 adrenergic agonist, ophthalmic · also: Apraclonidine

How it works: Stimulates alpha-2 receptors → decreases aqueous humor production and increases outflow → lowers IOP (intraocular pressure).

INDICATIONS
- Open-angle glaucoma
- Ocular hypertension

MAIN SIDE EFFECTS
- Ocular burning, stinging, itching; allergic conjunctivitis
- Dry mouth
- Drowsiness, fatigue, headache

ADVERSE EFFECTS
- Hypotension (systemic absorption)
- CNS depression and apnea in infants/young children

CONTRAINDICATIONS & PRECAUTIONS
- Concurrent MAOIs (monoamine oxidase inhibitors)
- Children under 2 years
- Caution: severe cardiovascular disease, depression

INTERACTIONS
- MAOIs → contraindicated
- CNS depressants, alcohol → additive sedation
- Antihypertensives → additive hypotension

SPECIAL WARNINGS
- Assess for drowsiness and allergic eye reaction (red, itchy eyes)

CLIENT EDUCATION
- Remove soft contact lenses; wait 15 min before reinserting
- May cause drowsiness — use caution driving
- Report red, itchy, swollen eyes

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Brimonidine works like clonidine for the eye — and is dangerous in infants (apnea).

Dorzolamide *(Trusopt; Cosopt (with timolol))*

Ch 13 · Glaucoma

Carbonic anhydrase inhibitor (CAI), topical ophthalmic · also: Brinzolamide

How it works: Inhibits carbonic anhydrase in the ciliary body → decreases aqueous humor production → lowers IOP (intraocular pressure).

INDICATIONS
- Open-angle glaucoma
- Ocular hypertension

MAIN SIDE EFFECTS
- Bitter taste after instilling
- Ocular burning, stinging, blurred vision
- Conjunctivitis, eyelid reactions

ADVERSE EFFECTS
- Sulfonamide-type hypersensitivity reactions (rare)

CONTRAINDICATIONS & PRECAUTIONS
- Sulfonamide (sulfa) allergy
- Severe renal impairment
- Caution: hepatic impairment

INTERACTIONS
- Oral carbonic anhydrase inhibitors (acetazolamide) → additive effects; avoid combination

SPECIAL WARNINGS
- Topical — far fewer systemic effects than oral acetazolamide

CLIENT EDUCATION
- A bitter taste after the drop is expected
- Remove soft contacts; wait 15 min before reinserting
- Report eye swelling, rash, or itching
- Wait at least 5 min between eye drops

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Dorzolamide is a sulfonamide — always ask about sulfa allergy.

Acetazolamide *(Diamox)*

Ch 13 · Glaucoma

Carbonic anhydrase inhibitor (CAI), systemic (oral/IV) · also: Methazolamide

How it works: Inhibits carbonic anhydrase → decreases aqueous humor production → lowers IOP; also a weak diuretic.

INDICATIONS
- Acute angle-closure glaucoma (emergency, to lower IOP before surgery)
- Open-angle glaucoma (adjunct)
- Also: altitude sickness, edema, seizures

MAIN SIDE EFFECTS
- Paresthesias (tingling of fingers, toes, lips)
- Anorexia, nausea, metallic taste
- Increased urination, drowsiness

ADVERSE EFFECTS
- Hypokalemia, metabolic acidosis — monitor electrolytes
- Kidney stones
- Blood dyscrasias, Stevens-Johnson syndrome (sulfonamide)

CONTRAINDICATIONS & PRECAUTIONS
- Sulfonamide (sulfa) allergy
- Hypokalemia, hyponatremia, hyperchloremic acidosis
- Severe renal or hepatic disease
- Caution: diabetes, COPD

INTERACTIONS
- High-dose aspirin → acetazolamide toxicity (lethargy, confusion, acidosis)
- Other diuretics, corticosteroids → increased hypokalemia
- Topical CAIs → additive; avoid combination

SPECIAL WARNINGS
- Monitor serum potassium, sodium, bicarbonate, CBC
- Monitor I&O and daily weight

CLIENT EDUCATION
- Tingling of fingers and toes is common
- Drink fluids to help prevent kidney stones
- Report rash, sore throat, unusual bleeding, flank pain
- Eat potassium-rich foods if prescribed

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Oral CAI for angle-closure emergency — sulfa allergy and hypokalemia are the test traps.

Pilocarpine (ophthalmic) *(Isopto Carpine)*

Ch 13 · Glaucoma

Cholinergic (muscarinic) agonist — miotic · also: Carbachol

How it works: Contracts pupil (miosis) and ciliary muscle → opens trabecular meshwork → increases aqueous outflow → lowers IOP (intraocular pressure).

INDICATIONS
- Open-angle glaucoma
- Acute angle-closure glaucoma (to constrict pupil)
- Reversal of drug-induced mydriasis

MAIN SIDE EFFECTS
- Blurred vision, nearsightedness (accommodation spasm)
- Poor vision in dim light / at night (small pupil)
- Brow ache, headache
- Eye stinging

ADVERSE EFFECTS
- Retinal detachment (rare) — report flashes, floaters, curtain over vision
- Systemic cholinergic toxicity: sweating, salivation, bradycardia, bronchospasm

CONTRAINDICATIONS & PRECAUTIONS
- Conditions where miosis is undesirable (acute iritis/uveitis)
- Caution: asthma, cardiac disease
- Caution: history of retinal detachment

INTERACTIONS
- Anticholinergic eye drops (atropine) → oppose effect

SPECIAL WARNINGS
- Assess for systemic cholinergic effects
- Blurred vision is worst right after the dose

CLIENT EDUCATION
- Don't drive at night or in dim light
- Headache/brow ache usually lessens with use
- Report flashes of light, floaters, or vision loss
- Use punctal occlusion to reduce systemic absorption

ANTIDOTE
- Atropine (for systemic cholinergic toxicity)

Remember: Pilocarpine = PINpoint pupils → poor night vision.

Mannitol (osmotic agent) *(Osmitrol)*

Ch 13 · Glaucoma

Osmotic diuretic

How it works: Raises plasma osmolality → pulls fluid out of the eye into blood → rapidly lowers IOP (intraocular pressure).

INDICATIONS
- Acute angle-closure glaucoma (emergency, IV)
- Lower IOP before eye surgery
- Also: increased intracranial pressure, oliguria

MAIN SIDE EFFECTS
- Headache, nausea, vomiting
- Thirst, dry mouth
- Polyuria

ADVERSE EFFECTS
- Fluid and electrolyte imbalance, dehydration — monitor I&O, electrolytes
- Fluid overload: heart failure, pulmonary edema — monitor lung sounds
- Extravasation → tissue necrosis

CONTRAINDICATIONS & PRECAUTIONS
- Anuria
- Severe dehydration
- Severe pulmonary congestion/edema, severe heart failure
- Active intracranial bleeding (except during craniotomy)

INTERACTIONS
- Other diuretics → additive fluid/electrolyte loss
- Lithium → increased lithium excretion

SPECIAL WARNINGS
- Use an in-line filter; if crystals present, warm to dissolve and cool to body temp first
- Acute angle-closure: sudden severe eye pain, headache, nausea/vomiting, halos, blurred vision — emergency
- Monitor renal function, BUN, serum osmolality

CLIENT EDUCATION
- Expect increased urination
- Report shortness of breath or chest tightness
- Acute angle-closure needs laser/surgical iridotomy after IOP is lowered

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Angle-closure emergency combo: mannitol IV + acetazolamide + pilocarpine, then laser iridotomy.

EYE EXAMS

Mydriatics & cycloplegics (atropine, cyclopentolate eye drops; phenylephrine) *(Cyclogyl (cyclopentolate), Isopto Atropine)*

Ch 13 · Eye exams

Anticholinergic mydriatic-cycloplegics; phenylephrine = alpha-1 adrenergic agonist mydriatic · also: Tropicamide, homatropine, scopolamine (anticholinergic)

How it works: Anticholinergics dilate pupil (mydriasis) and paralyze accommodation (cycloplegia); phenylephrine dilates without cycloplegia.

INDICATIONS
- Eye exams (retinal exam, refraction)
- Before/after eye surgery
- Uveitis — rests the eye and reduces pain (atropine)

MAIN SIDE EFFECTS
- Blurred vision, photophobia
- Local stinging

ADVERSE EFFECTS
- Systemic anticholinergic toxicity (infants, children, older adults): tachycardia, flushing, fever, dry mouth, confusion/delirium
- Phenylephrine (10%) → hypertension, tachycardia
- May trigger acute angle-closure glaucoma

CONTRAINDICATIONS & PRECAUTIONS
- Angle-closure glaucoma or narrow angles
- Caution: infants, young children, older adults
- Phenylephrine: Caution — hypertension, cardiac disease

INTERACTIONS
- Other anticholinergics → additive effects
- Phenylephrine + MAOIs or tricyclic antidepressants → hypertension

SPECIAL WARNINGS
- Atropine eye effects can last 1–2 weeks; cyclopentolate about 24 hr
- Use punctal occlusion, especially in children

CLIENT EDUCATION
- Wear sunglasses; light will be bothersome
- Don't drive until vision returns to normal
- Report eye pain, fever, fast heartbeat, or confusion

ANTIDOTE
- Physostigmine (for severe systemic anticholinergic toxicity)

Remember: Cyclo-plegic = ciliary paralysis (can't focus); mydriatic = dilated pupil. Never in angle-closure glaucoma.

EAR DISORDERS

Otic antibiotics & ear drop technique *(Cipro HC, Ciprodex, Cortisporin Otic)*

Ch 13 · Ear disorders

Topical otic antibacterials (often with a corticosteroid) · also: Ciprofloxacin otic, ofloxacin otic, neomycin/polymyxin B/hydrocortisone

How it works: Kill bacteria in the ear canal; added steroid reduces inflammation and swelling.

INDICATIONS
- Otitis externa (swimmer's ear)
- Otitis media with ear tubes or perforated eardrum (ofloxacin, ciprofloxacin/dexamethasone)

MAIN SIDE EFFECTS
- Ear discomfort, itching, stinging
- Dizziness/vertigo if drops are cold

ADVERSE EFFECTS
- Allergic reaction (rash, swelling)
- Superinfection (fungal) with prolonged use

CONTRAINDICATIONS & PRECAUTIONS
- Perforated eardrum: avoid ototoxic neomycin-containing drops unless prescribed
- Hydrocortisone-containing drops: viral ear infections (herpes, varicella)

SPECIAL WARNINGS
- Assess for drainage, pain, and hearing changes
- Complete the full course

CLIENT EDUCATION
- Warm bottle in hands first — cold drops cause dizziness/nausea
- Lie on side, affected ear up
- Adult (and child over 3): pull pinna UP and BACK
- Child under 3: pull pinna DOWN and BACK
- Stay on side about 5 min; don't touch dropper to ear
- Keep ear dry; avoid swimming as directed

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Under 3 = down and back (the ear canal is short and points up in young kids).

EYE EXAMS

Eye drop & ointment administration

Ch 13 · Eye exams

Administration technique

How it works: Correct technique keeps the drug in the eye and limits systemic absorption.

INDICATIONS
- Any ophthalmic drop or ointment

ADVERSE EFFECTS
- Systemic absorption via tear duct (e.g., beta blockers, anticholinergics)
- Contamination → eye infection if tip touches eye

SPECIAL WARNINGS
- Wait at least 5 min between different eye drops
- Drops first, ointment last
- Remove contacts unless told otherwise; wait 15 min to reinsert

CLIENT EDUCATION
- Wash hands; tilt head back and look up
- Pull lower lid down; drop into the conjunctival sac — not on the cornea
- Close eye gently (don't squeeze) and press inner corner 1–2 min
- Don't touch the dropper tip to the eye or anything else
- Ointment: thin ribbon inner to outer corner; vision blurs briefly

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Punctal occlusion (press inner corner) = more drug in eye, less in bloodstream.

Ch 14 · Muscle Relaxants (only)

CENTRALLY ACTING MUSCLE RELAXANTS

Baclofen *(Lioresal; Gablofen (intrathecal))*

Ch 14 · Centrally acting muscle relaxants

Centrally acting muscle relaxant (antispasmodic) · also: Cyclobenzaprine, tizanidine, diazepam, methocarbamol, carisoprodol

How it works: GABA (gamma-aminobutyric acid)-B agonist in the spinal cord; inhibits reflexes that cause spasticity.

INDICATIONS

- Spasticity from multiple sclerosis (MS) or spinal cord injury
- Severe spasticity (e.g., cerebral palsy) via intrathecal pump

MAIN SIDE EFFECTS

- Drowsiness, dizziness, fatigue
- Muscle weakness
- Nausea, constipation
- Urinary retention

ADVERSE EFFECTS

- CNS (central nervous system) depression / respiratory depression with overdose — monitor RR, LOC
- Abrupt withdrawal → hallucinations, seizures, high fever, rebound spasticity

CONTRAINDICATIONS & PRECAUTIONS

- Caution: seizure disorders (may lower seizure threshold)
- Caution: renal impairment (excreted by kidneys)
- Caution: older adults (sedation, falls)

INTERACTIONS

- Alcohol, opioids, benzodiazepines, other CNS depressants → additive sedation; avoid

SPECIAL WARNINGS

- **BLACK BOX:** abrupt stop of intrathecal baclofen → high fever, altered mental status, rebound spasticity, rhabdomyolysis, organ failure, death
- Intrathecal pump: keep refill appointments; know pump alarm
- Taper oral doses gradually (over 1–2 weeks) — never stop abruptly
- Assess spasticity and muscle tone for effectiveness

CLIENT EDUCATION

- Don't stop suddenly — taper with your provider
- Avoid alcohol and driving until you know how it affects you
- Change positions slowly; report weakness or confusion
- Never miss an intrathecal pump refill appointment

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Baclofen = intrathecal pump + the classic 'never stop abruptly' drug — withdrawal can kill.

Cyclobenzaprine *(Flexeril, Amrix)*

Ch 14 · Centrally acting muscle relaxants

Centrally acting muscle relaxant (structurally related to tricyclic antidepressants) · also: Baclofen, tizanidine, methocarbamol, carisoprodol

How it works: Acts in the brainstem to reduce motor activity; structurally similar to TCAs (tricyclic antidepressants).

INDICATIONS

- Acute, painful musculoskeletal spasm (short-term, 2–3 weeks)
- NOT effective for spasticity from cerebral palsy or spinal cord disease

MAIN SIDE EFFECTS

- Drowsiness, dizziness
- Anticholinergic: dry mouth, blurred vision, constipation, urinary retention

ADVERSE EFFECTS

- Dysrhythmias / tachycardia (TCA-like cardiac effects)
- Serotonin syndrome with serotonergic drugs — agitation, hyperthermia, tremor

CONTRAINDICATIONS & PRECAUTIONS

- Recent MI (myocardial infarction), dysrhythmias, heart block, heart failure
- MAOIs (monoamine oxidase inhibitors) within 14 days
- Hyperthyroidism
- Caution: older adults (Beers list — sedation, anticholinergic effects, falls)
- Caution: glaucoma, urinary retention / BPH (benign prostatic hyperplasia)

INTERACTIONS

- MAOIs → hyperpyretic crisis, seizures, death; contraindicated
- SSRIs, SNRIs, tramadol → serotonin syndrome risk
- Alcohol, other CNS depressants → additive sedation
- Other anticholinergics → additive anticholinergic effects

SPECIAL WARNINGS

- For short-term use only (2–3 weeks)
- Monitor for anticholinergic effects and fall risk
- Use with rest and physical therapy for muscle spasm

CLIENT EDUCATION

- Sugarless gum, sips of water for dry mouth
- Increase fluids and fiber for constipation
- Avoid alcohol and driving until effects are known
- Report urinary retention, fast heartbeat, or confusion

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Cyclobenzaprine acts like a TCA — think anticholinergic effects + heart concerns + no MAOIs.

Tizanidine *(Zanaflex)*

Ch 14 · Centrally acting muscle relaxants

Centrally acting alpha-2 adrenergic agonist muscle relaxant · also: Baclofen, cyclobenzaprine, diazepam

How it works: Stimulates central alpha-2 receptors → reduces nerve impulses that cause spasticity.

INDICATIONS
- Spasticity from MS (multiple sclerosis) or spinal cord injury

MAIN SIDE EFFECTS
- Drowsiness, sedation
- Dry mouth
- Dizziness, weakness, fatigue

ADVERSE EFFECTS
- Hypotension (orthostatic), bradycardia — monitor BP and HR
- Hepatotoxicity — monitor LFTs (liver function tests)
- Hallucinations / psychotic symptoms

CONTRAINDICATIONS & PRECAUTIONS
- Ciprofloxacin or fluvoxamine (strong CYP1A2 inhibitors) — contraindicated
- Caution: hepatic or renal impairment
- Caution: older adults

INTERACTIONS
- Ciprofloxacin, fluvoxamine (block CYP1A2) → large rise in tizanidine level → severe hypotension, sedation; contraindicated
- Antihypertensives → additive hypotension
- Alcohol, CNS depressants → additive sedation; alcohol also raises tizanidine level
- Oral contraceptives → raise tizanidine level; dose may be lowered

SPECIAL WARNINGS
- LFTs at baseline and periodically during therapy
- Taper to stop — abrupt stop → rebound hypertension, tachycardia, increased spasticity
- Take consistently with or without food (food changes absorption)

CLIENT EDUCATION
- Rise slowly from sitting or lying down
- Report yellow skin/eyes, dark urine, nausea, or fatigue (liver)
- Avoid alcohol
- Do not stop suddenly

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Tizanidine is clonidine's cousin (alpha-2 agonist) → hypotension; plus liver + no cipro/fluvoxamine.

Diazepam (as a muscle relaxant)

Ch 14 · Centrally acting muscle relaxants

(Valium)

Benzodiazepine — centrally acting muscle relaxant · also: Baclofen, cyclobenzaprine, tizanidine

How it works: Enhances GABA (gamma-aminobutyric acid) inhibition in the CNS, including spinal reflexes → reduces muscle spasm.

INDICATIONS
- Muscle spasm and spasticity (e.g., cerebral palsy, spinal cord injury)
- Also: anxiety, alcohol withdrawal, seizures/status epilepticus, preprocedure sedation

MAIN SIDE EFFECTS
- Drowsiness, sedation
- Ataxia, dizziness
- Anterograde amnesia

ADVERSE EFFECTS
- Respiratory depression (especially IV, or with alcohol/opioids) — monitor RR, SpO2
- Dependence; withdrawal if stopped abruptly (anxiety, tremor, seizures)
- Paradoxical excitement (agitation), especially older adults/children

CONTRAINDICATIONS & PRECAUTIONS
- Acute angle-closure glaucoma
- Severe respiratory insufficiency, sleep apnea
- Myasthenia gravis
- Severe hepatic impairment
- Pregnancy
- Caution: older adults (falls), history of substance use disorder

INTERACTIONS
- Opioids → profound sedation, respiratory depression, coma, death
- Alcohol, other CNS depressants → additive CNS depression
- Grapefruit juice, cimetidine → raise diazepam level

SPECIAL WARNINGS
- **BLACK BOX:** concurrent opioids → respiratory depression, coma, death
- **BLACK BOX:** abuse, misuse, addiction; dependence and withdrawal reactions
- Schedule IV controlled substance
- IV: give slowly; have flumazenil and airway equipment ready

CLIENT EDUCATION
- Avoid alcohol and driving
- Do not stop suddenly — taper
- Report excessive drowsiness or slow breathing
- Store safely; risk of misuse

ANTIDOTE
- Flumazenil

Remember: Benzodiazepine overdose → flumazenil; flumazenil can trigger seizures in long-term benzo users.

Methocarbamol & carisoprodol (other centrally acting agents) *(Robaxin (methocarbamol), Soma (carisoprodol))*

Ch 14 · Centrally acting muscle relaxants

Centrally acting muscle relaxants · also: Metaxalone, chlorzoxazone, orphenadrine

How it works: General CNS depression → reduces acute muscle spasm; no direct effect on muscle.

INDICATIONS
- Acute, painful musculoskeletal conditions (short-term, 2–3 weeks)

MAIN SIDE EFFECTS
- Drowsiness, dizziness
- Headache, nausea
- Methocarbamol: urine may turn brown, black, or green (harmless)

ADVERSE EFFECTS
- Carisoprodol: abuse, dependence, and withdrawal (metabolized to meprobamate)
- CNS depression with other depressants

CONTRAINDICATIONS & PRECAUTIONS
- Carisoprodol: Caution — history of substance use disorder
- Caution: hepatic or renal impairment
- Caution: older adults (sedation, falls)

INTERACTIONS
- Alcohol, opioids, benzodiazepines, other CNS depressants → additive sedation; avoid

SPECIAL WARNINGS
- Carisoprodol is a Schedule IV controlled substance
- Short-term use only
- Assess fall risk and sedation

CLIENT EDUCATION
- Methocarbamol may darken urine — this is expected
- Avoid alcohol and driving until effects are known
- Carisoprodol: take only as prescribed; don't stop suddenly after long use

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Methocarbamol = funny-colored urine; carisoprodol = abuse potential (controlled).

PERIPHERALLY ACTING MUSCLE RELAXANTS

Dantrolene *(Dantrium, Ryanodex)*

Ch 14 · Peripherally acting muscle relaxants

Peripherally (direct) acting skeletal muscle relaxant

How it works: Acts directly on skeletal muscle — blocks calcium release from sarcoplasmic reticulum → weaker contraction.

INDICATIONS
- Spasticity (multiple sclerosis, spinal cord injury, cerebral palsy, stroke)
- Malignant hyperthermia (IV) — treatment and prevention

MAIN SIDE EFFECTS
- Drowsiness, dizziness
- Muscle weakness
- Diarrhea
- Fatigue, malaise

ADVERSE EFFECTS
- Hepatotoxicity — monitor LFTs; report jaundice, dark urine, abdominal pain
- Photosensitivity
- Excessive muscle weakness

CONTRAINDICATIONS & PRECAUTIONS
- Active liver disease (oral)
- Caution: women and clients over 35 (higher hepatotoxicity risk)
- Caution: cardiac or pulmonary impairment

INTERACTIONS
- Alcohol, CNS depressants → additive CNS depression
- Estrogens → increased hepatotoxicity risk
- IV verapamil (calcium channel blockers) with IV dantrolene → hyperkalemia, cardiovascular collapse; avoid

SPECIAL WARNINGS
- **BLACK BOX:** hepatotoxicity (oral) — LFTs at baseline and periodically
- Malignant hyperthermia: triggered by succinylcholine/volatile anesthetics → rigidity, rising temp, tachycardia, high end-tidal CO2
- MH: stop trigger, give IV dantrolene rapidly, cool the client
- Reconstitute IV dantrolene with sterile water (no bacteriostatic water)

CLIENT EDUCATION
- Wear sunscreen and protective clothing
- Report yellow skin/eyes, dark urine, abdominal pain
- Avoid alcohol and driving until effects are known

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Dantrolene IS the treatment for malignant hyperthermia — know it acts on muscle, not the CNS.

GENERAL

Muscle relaxants — general client education

Ch 14 · General

Centrally and peripherally acting muscle relaxants

How it works: Centrally acting agents depress the CNS; dantrolene acts directly on muscle.

INDICATIONS

- Acute muscle spasm (short-term agents)
- Spasticity from multiple sclerosis, spinal cord injury, cerebral palsy

MAIN SIDE EFFECTS

- Drowsiness, dizziness
- Muscle weakness

ADVERSE EFFECTS

- CNS depression, respiratory depression with other depressants
- Hepatotoxicity (tizanidine, dantrolene) — monitor LFTs

CONTRAINDICATIONS & PRECAUTIONS

- Caution: older adults — fall risk
- Caution: hepatic impairment

INTERACTIONS

- Alcohol, opioids, benzodiazepines, antihistamines → additive CNS depression

SPECIAL WARNINGS

- Assess fall risk and implement safety measures
- Spasticity agents (baclofen, tizanidine, diazepam): taper, do not stop abruptly

CLIENT EDUCATION

- Avoid alcohol and other CNS depressants
- Don't drive or use machinery until you know the effect
- Don't stop the medication suddenly — talk to your provider
- Change positions slowly
- Combine with rest, heat/ice, and physical therapy as prescribed

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Spasm (acute injury) = short-term agents like cyclobenzaprine; spasticity (multiple sclerosis, spinal cord injury, cerebral palsy) = baclofen, tizanidine, dantrolene, diazepam.

Ch 16 · Airflow Disorders (asthma/COPD)

BRONCHODILATORS: BETA2-ADRENERGIC AGONISTS

Albuterol *(Proventil, Ventolin, ProAir)*

Ch 16 · Bronchodilators: beta2-adrenergic agonists

Short-acting beta2-adrenergic agonist (SABA), bronchodilator · also: Levalbuterol

How it works: Stimulates beta2 receptors in bronchial smooth muscle → bronchodilation within minutes.

INDICATIONS
- Rescue for acute asthma attacks / acute bronchospasm
- Prevention of exercise-induced bronchospasm (inhale before exercise)
- COPD (chronic obstructive pulmonary disease) symptom relief

MAIN SIDE EFFECTS
- Tremor, nervousness, restlessness
- Tachycardia, palpitations (more with oral tablets)
- Headache

ADVERSE EFFECTS
- Angina / dysrhythmias with excess use — report chest pain, monitor HR
- Hypokalemia at high doses — monitor potassium

CONTRAINDICATIONS & PRECAUTIONS
- Caution: cardiac disease, dysrhythmias
- Caution: hyperthyroidism
- Caution: diabetes mellitus (can raise glucose)
- Caution: seizure disorders

INTERACTIONS
- Nonselective beta blockers (propranolol) → block bronchodilation; avoid
- MAOIs (monoamine oxidase inhibitors) / tricyclic antidepressants → cardiovascular toxicity; caution

SPECIAL WARNINGS
- Rescue drug — fast onset; NOT a long-term controller
- Needing it more than usual = asthma worsening; reassess therapy
- Monitor HR, lung sounds, peak flow, oxygen saturation

CLIENT EDUCATION
- Carry your rescue inhaler at all times
- Call provider if you need it more often or it stops working
- Report chest pain or racing heart
- Inhaled form has fewer side effects than tablets

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: SABA = rescue. 'Albuterol is for ALL-of-a-sudden.' Use it before any other inhaler.

Salmeterol *(Serevent (with fluticasone: Advair))*

Ch 16 · Bronchodilators: beta2-adrenergic agonists

Long-acting beta2-adrenergic agonist (LABA), bronchodilator · also: Formoterol

How it works: Long-acting beta2 stimulation → sustained bronchodilation (about 12 hr); slow onset.

INDICATIONS
- Long-term control of asthma (with an inhaled corticosteroid)
- Maintenance therapy for COPD (chronic obstructive pulmonary disease)
- Prevention of exercise-induced bronchospasm

MAIN SIDE EFFECTS
- Tremor, nervousness
- Tachycardia, palpitations
- Headache

ADVERSE EFFECTS
- Severe asthma exacerbation / asthma-related death (see black box)
- Paradoxical bronchospasm — stop and use rescue inhaler

CONTRAINDICATIONS & PRECAUTIONS
- Asthma treatment WITHOUT an inhaled corticosteroid
- Acute bronchospasm (not a rescue drug)
- Caution: cardiac disease, dysrhythmias, hyperthyroidism, diabetes

INTERACTIONS
- Nonselective beta blockers → reduced bronchodilation; avoid
- MAOIs / tricyclic antidepressants → cardiovascular toxicity; caution

SPECIAL WARNINGS
- **BLACK BOX:** asthma-related death — in asthma, use only with an inhaled corticosteroid
- NEVER use for an acute attack — onset too slow
- Twice daily dosing, about 12 hr apart

CLIENT EDUCATION
- This is a controller, not a rescue inhaler — keep albuterol handy
- Use every day as prescribed, even when you feel well
- Do not use more than twice a day

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: LABA alone in asthma = black box death risk. Always pair with an inhaled steroid (e.g. Advair = fluticasone + salmeterol).

BRONCHODILATORS: ANTICHOLINERGICS

Ipratropium *(Atrovent (with albuterol: Combivent, DuoNeb))*

Ch 16 · Bronchodilators: anticholinergics

Short-acting inhaled anticholinergic (muscarinic antagonist), bronchodilator · also: Tiotropium (long-acting)

How it works: Blocks muscarinic (acetylcholine) receptors in the airways → bronchodilation.

INDICATIONS
- COPD (chronic obstructive pulmonary disease) — bronchospasm
- Asthma (off-label / add-on)
- Rhinitis (intranasal form)

MAIN SIDE EFFECTS
- Dry mouth
- Hoarseness, throat irritation
- Headache, cough

ADVERSE EFFECTS
- Paradoxical bronchospasm — rare
- Urinary retention, worsening glaucoma — monitor

CONTRAINDICATIONS & PRECAUTIONS
- Allergy to peanuts or soy (older MDI formulations contained soy lecithin)
- Caution: narrow-angle glaucoma
- Caution: BPH (benign prostatic hyperplasia) / urinary retention

INTERACTIONS
- Other anticholinergics → additive anticholinergic effects

SPECIAL WARNINGS
- Ask about peanut/soy allergy before giving
- Not ideal as sole rescue drug — slower onset than albuterol
- Monitor I&O / urinary retention in older men

CLIENT EDUCATION
- Sip water, suck sugarless hard candy for dry mouth
- Avoid spraying in eyes — can blur vision / worsen glaucoma
- Report trouble urinating or eye pain

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Classic NCLEX: check peanut/soy allergy before ipratropium. Anticholinergic → 'dry as a bone' (dry mouth).

Tiotropium *(Spiriva HandiHaler, Spiriva Respimat)*

Ch 16 · Bronchodilators: anticholinergics

Long-acting inhaled anticholinergic (muscarinic antagonist), bronchodilator · also: Ipratropium (short-acting)

How it works: Long-acting blockade of airway muscarinic receptors → sustained bronchodilation; once daily.

INDICATIONS
- Maintenance therapy for COPD (chronic obstructive pulmonary disease)
- Asthma maintenance (Respimat, add-on)

MAIN SIDE EFFECTS
- Dry mouth
- Constipation
- Upper respiratory infection symptoms, sinusitis

ADVERSE EFFECTS
- Urinary retention
- Worsening narrow-angle glaucoma / eye pain if powder gets in eyes
- Paradoxical bronchospasm

CONTRAINDICATIONS & PRECAUTIONS
- Allergy to atropine or ipratropium
- Caution: narrow-angle glaucoma
- Caution: BPH (benign prostatic hyperplasia) / bladder neck obstruction

INTERACTIONS
- Other anticholinergics → additive effects; avoid combining

SPECIAL WARNINGS
- Maintenance only — NOT for acute bronchospasm
- Once daily dosing

CLIENT EDUCATION
- HandiHaler: put capsule in chamber, press button to pierce, inhale deeply
- Never swallow the capsule
- Do not get powder in your eyes
- Use at the same time each day

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Spiriva capsule goes IN the HandiHaler, never in the mouth to swallow.

BRONCHODILATORS: METHYLXANTHINES

Theophylline *(Theo-24, Elixophyllin)*

Ch 16 · Bronchodilators: methylxanthines

Methylxanthine bronchodilator · also: Aminophylline (IV)

How it works: Relaxes bronchial smooth muscle → bronchodilation; also CNS and cardiac stimulation (caffeine relative).

INDICATIONS
- Long-term control of chronic asthma (not first-line)
- COPD (chronic obstructive pulmonary disease)

MAIN SIDE EFFECTS
- Nausea, vomiting, GI upset
- Restlessness, nervousness, insomnia
- Headache

ADVERSE EFFECTS
- Toxicity >20 mcg/mL: N/V, diarrhea, insomnia, restlessness
- Higher levels: dysrhythmias (tachycardia, ventricular fibrillation), seizures
- Treat toxicity: stop drug, activated charcoal, lidocaine for dysrhythmias, diazepam for seizures

CONTRAINDICATIONS & PRECAUTIONS
- Uncontrolled seizure disorder or dysrhythmias
- Caution: cardiac disease, peptic ulcer, hyperthyroidism
- Caution: liver disease (slower clearance)

INTERACTIONS
- Caffeine → additive CNS stimulation; limit coffee, tea, cola, chocolate
- Cimetidine, ciprofloxacin (fluoroquinolones) → raise theophylline level; toxicity
- Phenobarbital, phenytoin, rifampin → lower theophylline level
- Smoking → lowers level; quitting raises it — dose may need adjusting

SPECIAL WARNINGS
- Therapeutic level 10–20 mcg/mL — narrow therapeutic range
- Monitor serum levels, HR/rhythm, signs of toxicity
- Not a rescue drug

CLIENT EDUCATION
- Avoid caffeine (coffee, tea, energy drinks, chocolate)
- Tell provider if you start or stop smoking
- Report vomiting, fast heartbeat, restlessness — signs of toxicity
- Keep blood level appointments; do not crush extended-release forms

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Theophylline 10–20. Smoking speeds its breakdown → stopping smoking can push the level toxic.

Fluticasone (inhaled)

(Flovent)

Inhaled corticosteroid (ICS), glucocorticoid · also: Beclomethasone, budesonide, mometasone

How it works: Reduces airway inflammation and swelling, decreases mucus production; long-term control.

INDICATIONS
- Long-term prophylaxis/control of asthma
- COPD (chronic obstructive pulmonary disease) with frequent exacerbations

MAIN SIDE EFFECTS
- Oropharyngeal candidiasis (thrush)
- Dysphonia (hoarseness)
- Throat irritation, cough

ADVERSE EFFECTS
- High doses / long-term: adrenal suppression
- Bone loss (osteoporosis)
- Slowed growth in children — monitor height
- Hyperglycemia — monitor glucose

CONTRAINDICATIONS & PRECAUTIONS
- Acute bronchospasm / status asthmaticus (not a rescue drug)
- Caution: active infections (fungal, TB)
- Caution: diabetes, osteoporosis

SPECIAL WARNINGS
- Controller only — does not relieve an acute attack
- Full benefit takes days to weeks of regular use
- Treat thrush with antifungal (e.g. nystatin) if it develops

CLIENT EDUCATION
- Rinse mouth with water and spit after each use
- Use a spacer with MDI to reduce thrush and hoarseness
- If also using a bronchodilator, use it first, wait 5 min, then steroid
- Use daily even when symptom-free; report white patches in mouth

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Steroid inhaler: Rinse, Spit, Spacer. Bronchodilator opens the door first, steroid goes in second.

Prednisone (oral)

Oral glucocorticoid (systemic corticosteroid) · also: Prednisolone, methylprednisolone (IV/oral)

How it works: Systemic suppression of inflammation and immune response → reduces airway inflammation.

INDICATIONS
- Short-term treatment of acute asthma / COPD exacerbation
- Severe persistent asthma not controlled by other drugs

MAIN SIDE EFFECTS
- Increased appetite, weight gain, fluid retention
- Mood changes, insomnia
- GI upset

ADVERSE EFFECTS
- Adrenal suppression — taper; never stop abruptly
- Infection (masks signs) — monitor temp, WBC, wounds
- Hyperglycemia — monitor glucose
- Osteoporosis (long-term)
- Peptic ulcer / GI bleeding
- Hypokalemia, fluid retention

CONTRAINDICATIONS & PRECAUTIONS
- Systemic fungal infections
- Live virus vaccines
- Caution: diabetes, peptic ulcer disease, heart failure, hypertension

INTERACTIONS
- NSAIDs → GI ulcers/bleeding
- Potassium-depleting diuretics → hypokalemia
- Digoxin → toxicity if hypokalemic
- Insulin / oral antidiabetics → may need dose increase

SPECIAL WARNINGS
- Taper after long-term use — risk of adrenal crisis if stopped suddenly
- Monitor glucose, potassium, weight, BP, signs of infection
- Long-term: bone density, eye exams (cataracts, glaucoma)

CLIENT EDUCATION
- Take with food, in the morning
- Do not stop suddenly — follow the taper
- Avoid crowds/sick people; report fever or sore throat
- Calcium + vitamin D, weight-bearing exercise for bones; report black stools

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Short burst for flares; taper if long-term. Steroids raise glucose and hide infections.

LEUKOTRIENE MODIFIERS

Montelukast *(Singulair)*

Ch 16 · Leukotriene modifiers

Leukotriene receptor antagonist · also: Zafirlukast, zileuton

How it works: Blocks leukotriene receptors → less bronchoconstriction, airway inflammation, edema, and mucus.

INDICATIONS
- Long-term control of asthma (age 12 months and up)
- Prevention of exercise-induced bronchospasm
- Allergic rhinitis

MAIN SIDE EFFECTS
- Headache
- GI upset
- Upper respiratory infection symptoms

ADVERSE EFFECTS
- Neuropsychiatric events: agitation, depression, sleep disturbances, suicidal thoughts
- Zafirlukast/zileuton: liver injury — monitor liver enzymes

CONTRAINDICATIONS & PRECAUTIONS
- Acute asthma attack (not a rescue drug)
- Caution: history of depression or psychiatric illness

INTERACTIONS
- Phenobarbital → lowers montelukast level
- Zafirlukast + warfarin → increased INR / bleeding (zafirlukast specific)

SPECIAL WARNINGS
- **BLACK BOX:** serious neuropsychiatric events (including suicidal thoughts/actions)
- Prophylaxis only — does not stop an attack in progress
- Zafirlukast/zileuton: monitor liver function

CLIENT EDUCATION
- Take once daily in the evening
- For exercise-induced bronchospasm: take at least 2 hr before exercise
- Report mood changes, nightmares, or suicidal thoughts right away
- Keep rescue inhaler for attacks

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Montelukast = evening pill + watch mood (black box neuropsych).

MONOCLONAL ANTIBODIES (BIOLOGICS)

Omalizumab *(Xolair)*

Ch 16 · Monoclonal antibodies (biologics)

Anti-IgE (immunoglobulin E) monoclonal antibody

How it works: Binds free IgE → prevents it from activating mast cells and basophils → fewer allergic asthma triggers.

INDICATIONS
- Moderate to severe persistent allergic asthma not controlled by inhaled corticosteroids
- Chronic idiopathic urticaria

MAIN SIDE EFFECTS
- Injection-site reactions (pain, bruising, redness)
- Upper respiratory infection, sinusitis
- Headache, joint pain

ADVERSE EFFECTS
- Anaphylaxis — can occur after any dose, even after long use
- Possible increased cancer risk — monitor

CONTRAINDICATIONS & PRECAUTIONS
- Hypersensitivity to omalizumab
- Acute bronchospasm (not a rescue drug)

SPECIAL WARNINGS
- **BLACK BOX:** anaphylaxis — give in a setting prepared to treat it
- Observe client after injection; have epinephrine available
- Subcutaneous injection every 2–4 weeks

CLIENT EDUCATION
- Know signs of anaphylaxis: wheezing, throat tightness, hives, fainting
- Carry an epinephrine auto-injector if prescribed
- Not for sudden attacks — keep rescue inhaler
- Do not stop other asthma meds unless told

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: '-mab' = monoclonal antibody. Xolair black box = anaphylaxis → monitor after every injection.

MAST CELL STABILIZERS

Cromolyn *(NasalCrom (nasal))*

Ch 16 · Mast cell stabilizers

Mast cell stabilizer, anti-inflammatory

How it works: Stabilizes mast cells → prevents release of histamine and other inflammatory mediators.

INDICATIONS
- Prophylaxis of asthma (nebulizer)
- Prevention of exercise- or allergen-induced bronchospasm
- Allergic rhinitis (nasal spray)

MAIN SIDE EFFECTS
- Unpleasant taste
- Throat irritation, cough
- Nasal stinging/sneezing (nasal form)

ADVERSE EFFECTS
- Rare: bronchospasm, hypersensitivity

CONTRAINDICATIONS & PRECAUTIONS
- Acute asthma attack — prophylaxis only

SPECIAL WARNINGS
- Prevention only — never for an acute attack
- Must be used regularly; full benefit may take several weeks
- Very safe — few systemic effects

CLIENT EDUCATION
- Use shortly before exercise or exposure to known allergen
- Use on schedule even when well
- Keep rescue inhaler for attacks

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Cromolyn is a shield, not a cure — prevents attacks but does nothing once one starts.

CLIENT EDUCATION: INHALER TECHNIQUE

Inhaler technique

Ch 16 · Client education: inhaler technique

Teaching: MDI (metered-dose inhaler), DPI (dry-powder inhaler), peak flow

How it works: Correct technique gets the drug to the lungs instead of the mouth and throat.

INDICATIONS
- Any client using inhaled asthma/COPD (chronic obstructive pulmonary disease) medications

SPECIAL WARNINGS
- Order: bronchodilator first, wait about 5 min, then corticosteroid
- Peak flow GREEN 80–100% of personal best: continue usual meds
- Peak flow YELLOW 50–79%: caution — use rescue, follow action plan
- Peak flow RED <50%: medical alert — rescue inhaler, seek care now

CLIENT EDUCATION
- MDI: shake, exhale fully, inhale slowly and deeply as you press, hold breath 10 sec
- Wait about 1 min between puffs
- Use a spacer if hand-breath coordination is hard (also less thrush)
- DPI: do NOT shake or exhale into it; inhale fast and deep, hold breath
- Peak flow: stand, take best of 3 tries, record daily

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: MDI = slow and deep. DPI = fast and deep. Opener (bronchodilator) before steroid.

ANTITUSSIVES

Codeine (antitussive)

Ch 17 · Antitussives

Opioid antitussive · also: Hydrocodone

How it works: Acts on the cough center in the medulla to suppress the cough reflex.

INDICATIONS
- Nonproductive cough (often combined with guaifenesin)

MAIN SIDE EFFECTS
- Drowsiness, dizziness
- Constipation
- Nausea/vomiting

ADVERSE EFFECTS
- Respiratory depression — check RR, hold if low, give naloxone
- Abuse / dependence potential (controlled substance)

CONTRAINDICATIONS & PRECAUTIONS
- Children under 12; opioid cough products are for adults 18 and older
- Caution: asthma, COPD (chronic obstructive pulmonary disease), emphysema
- Caution: head injury, history of substance use disorder

INTERACTIONS
- Alcohol, benzodiazepines, other CNS depressants → profound sedation, respiratory depression; avoid

SPECIAL WARNINGS
- Monitor respiratory rate and level of sedation
- Avoid suppressing a productive cough — secretions need clearing
- Some people are ultra-rapid metabolizers → dangerous morphine levels

CLIENT EDUCATION
- Avoid alcohol and other sedating drugs
- Do not drive until you know how it affects you
- Increase fluids and fiber to prevent constipation
- Store safely away from children

ANTIDOTE
- Naloxone

Remember: Opioid = respiratory depression → naloxone. Don't suppress a productive cough.

Dextromethorphan *(Robitussin DM, Delsym)*

Ch 17 · Antitussives

Non-opioid antitussive

How it works: Suppresses the cough center in the medulla; no analgesia, little respiratory depression at usual doses.

INDICATIONS
- Nonproductive cough

MAIN SIDE EFFECTS
- Drowsiness, dizziness
- Nausea, GI upset

ADVERSE EFFECTS
- Serotonin syndrome with serotonergic drugs (agitation, fever, tremor, tachycardia)
- Misuse at high doses → hallucinations, CNS depression

CONTRAINDICATIONS & PRECAUTIONS
- Use of MAOIs (monoamine oxidase inhibitors) within the past 14 days
- Caution: productive cough

INTERACTIONS
- MAOIs → serotonin syndrome / hyperpyrexia; contraindicated
- SSRIs (selective serotonin reuptake inhibitors) → serotonin syndrome risk; caution
- Alcohol / CNS depressants → additive sedation

SPECIAL WARNINGS
- Screen for MAOI and SSRI use before giving
- Abuse potential in teens — high-dose misuse

CLIENT EDUCATION
- Check labels — many cold products contain dextromethorphan; avoid double-dosing
- Tell provider if you take antidepressants
- Report cough lasting more than 1 week or with fever

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: 'DM' on the label = dextromethorphan. DM + MAOI = serotonin syndrome.

Benzonatate *(Tessalon Perles)*

Ch 17 · Antitussives

Non-opioid antitussive (local anesthetic action)

How it works: Numbs (anesthetizes) stretch receptors in the respiratory tract → less cough reflex.

INDICATIONS
- Nonproductive cough

MAIN SIDE EFFECTS
- Drowsiness, dizziness
- Headache
- GI upset, constipation

ADVERSE EFFECTS
- Chewing/sucking capsule → numb mouth and throat → choking, aspiration
- Accidental ingestion in children under 10 → fatal overdose (seizures, cardiac arrest)
- Hypersensitivity reactions: bronchospasm, laryngospasm

CONTRAINDICATIONS & PRECAUTIONS
- Allergy to ester-type local anesthetics (procaine, tetracaine)
- Not for children under 10

INTERACTIONS
- CNS depressants → additive sedation

SPECIAL WARNINGS
- Capsules look like candy — store out of children's reach
- Do not give if client cannot swallow capsules whole

CLIENT EDUCATION
- Swallow capsules whole — never chew, suck, or break them
- If mouth goes numb, do not eat or drink until feeling returns
- Keep locked away from children

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Benzonatate = 'numbing pearl': swallow whole or it numbs the throat → choking.

EXPECTORANTS

Guaifenesin *(Mucinex, Robitussin)*

Ch 17 · Expectorants

Expectorant

How it works: Thins and loosens respiratory secretions so cough becomes more productive.

INDICATIONS
- Productive cough with thick mucus (colds, bronchitis)

MAIN SIDE EFFECTS
- Nausea, vomiting, GI upset
- Drowsiness, dizziness
- Headache

CONTRAINDICATIONS & PRECAUTIONS
- Caution: persistent or chronic cough (smoking, asthma, emphysema) — needs evaluation

SPECIAL WARNINGS
- Often combined with antitussives or decongestants — check combination products

CLIENT EDUCATION
- Drink plenty of fluids to help thin mucus
- Report cough lasting more than 1 week, fever, rash, or persistent headache
- Do not crush or chew extended-release tablets

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Expectorant = helps you expectorate (cough it up). Fluids make it work better.

MUCOLYTICS

Acetylcysteine *(Mucomyst (inhaled), Acetadote (IV))*

Ch 17 · Mucolytics

Mucolytic; antidote for acetaminophen overdose · also: Dornase alfa (cystic fibrosis)

How it works: Breaks apart mucus bonds → thinner secretions; also restores liver glutathione after acetaminophen overdose.

INDICATIONS
- Thick secretions: COPD (chronic obstructive pulmonary disease), cystic fibrosis, pneumonia
- Acetaminophen overdose (oral or IV) — prevents liver damage

MAIN SIDE EFFECTS
- Unpleasant rotten-egg (sulfur) odor and taste
- Nausea, vomiting
- Stomatitis, rhinorrhea

ADVERSE EFFECTS
- Bronchospasm (inhaled) — especially in asthma; have bronchodilator ready
- Large volume of liquefied secretions — have suction available

CONTRAINDICATIONS & PRECAUTIONS
- Caution: asthma (bronchospasm risk)

SPECIAL WARNINGS
- Inhaled: give a bronchodilator if bronchospasm occurs; suction ready
- Acetaminophen overdose: give as soon as possible
- Oral dose for overdose: dilute in juice or soda to mask taste

CLIENT EDUCATION
- The smell is expected and fades
- Cough up loosened secretions; report wheezing or chest tightness

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Acetylcysteine does double duty: mucolytic AND the antidote for acetaminophen overdose.

DECONGESTANTS

Phenylephrine *(Neo-Synephrine, Sudafed PE)*

Ch 17 · Decongestants

Nasal decongestant (alpha1-adrenergic agonist) · also: Pseudoephedrine (oral), oxymetazoline (intranasal)

How it works: Stimulates alpha1 receptors → constricts nasal blood vessels → less swelling and congestion.

INDICATIONS
- Nasal congestion: allergic rhinitis, sinusitis, common cold

MAIN SIDE EFFECTS
- Nasal burning, stinging, dryness (intranasal)
- Restlessness, nervousness, insomnia (oral)

ADVERSE EFFECTS
- Rebound congestion with intranasal use longer than 3–5 days
- Systemic vasoconstriction → hypertension, dysrhythmias, stroke risk (mainly oral)

CONTRAINDICATIONS & PRECAUTIONS
- MAOI (monoamine oxidase inhibitor) use
- Caution: hypertension, coronary artery disease, dysrhythmias
- Caution: hyperthyroidism, diabetes, glaucoma, BPH (benign prostatic hyperplasia)

INTERACTIONS
- MAOIs → hypertensive crisis; avoid
- Other sympathomimetics (stimulants) → additive cardiovascular effects

SPECIAL WARNINGS
- Intranasal: limit to 3–5 days to prevent rebound congestion
- Monitor BP and HR in at-risk clients
- Pseudoephedrine kept behind the counter (methamphetamine precursor)

CLIENT EDUCATION
- Do not use nasal sprays longer than 3–5 days
- Check BP if you have hypertension; ask before using if heart disease
- Avoid caffeine and other stimulants
- Take oral forms early in the day to avoid insomnia

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Decongestant overuse → rebound congestion. Stop after 3–5 days.

ANTIHISTAMINES

Diphenhydramine *(Benadryl)*

Ch 17 · Antihistamines

First-generation H1 antihistamine · also: Chlorpheniramine, promethazine

How it works: Blocks H1 histamine receptors; crosses the blood-brain barrier → sedation; strong anticholinergic effects.

INDICATIONS
- Allergic rhinitis, mild allergic reactions (hives, itching)
- Adjunct in anaphylaxis (after epinephrine)
- Motion sickness
- Insomnia (short-term)

MAIN SIDE EFFECTS
- Sedation, drowsiness
- Anticholinergic: dry mouth, blurred vision, constipation, urinary retention
- Dizziness, GI upset

ADVERSE EFFECTS
- Paradoxical excitation in young children (restlessness, insomnia, seizures in overdose)
- Older adults: confusion, falls, delirium — on Beers list, avoid
- Overdose: severe anticholinergic toxicity

CONTRAINDICATIONS & PRECAUTIONS
- Newborns / infants and breastfeeding
- Caution: narrow-angle glaucoma
- Caution: BPH (benign prostatic hyperplasia) / urinary retention
- Caution: older adults

INTERACTIONS
- Alcohol, opioids, benzodiazepines → additive CNS depression
- Other anticholinergics → additive anticholinergic effects
- MAOIs (monoamine oxidase inhibitors) → intensified anticholinergic effects

SPECIAL WARNINGS
- Implement fall precautions in older adults
- Monitor urine output in clients with BPH

CLIENT EDUCATION
- Do not drive or operate machinery until you know the effect
- Avoid alcohol
- Sip water / sugarless candy for dry mouth
- Report trouble urinating

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: First-gen = sedating + anticholinergic. Kids may get wired instead of sleepy; older adults get confused and fall.

Loratadine *(Claritin)*

Ch 17 · Antihistamines

Second-generation H1 antihistamine (less sedating) · also: Cetirizine, fexofenadine, desloratadine

How it works: Blocks peripheral H1 histamine receptors; crosses blood-brain barrier poorly → little sedation.

INDICATIONS
- Seasonal and perennial allergic rhinitis
- Chronic urticaria (hives)

MAIN SIDE EFFECTS
- Headache
- Mild drowsiness (cetirizine most likely of the group)
- Dry mouth

ADVERSE EFFECTS
- Few serious effects — much less anticholinergic than first generation

CONTRAINDICATIONS & PRECAUTIONS
- Caution: kidney or liver impairment (dose adjustment)

INTERACTIONS
- Fexofenadine + fruit juice (apple, orange, grapefruit) → less absorption; take with water
- Fexofenadine + aluminum/magnesium antacids → less absorption; separate doses
- Alcohol / CNS depressants → additive drowsiness (mainly cetirizine)

SPECIAL WARNINGS
- Preferred over first-generation for older adults and daytime use
- Do not relieve nasal congestion — that needs a decongestant

CLIENT EDUCATION
- Take fexofenadine with water, not fruit juice
- Use cautiously when driving until you know how it affects you
- Works best taken regularly during allergy season

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Fexofenadine + fruit juice = poor absorption. Cetirizine is the 'sleepiest' of the second gens.

INTRANASAL GLUCOCORTICOIDS

Fluticasone (nasal) *(Flonase)*

Ch 17 · Intranasal glucocorticoids

Intranasal corticosteroid (glucocorticoid) · also: Budesonide, mometasone, triamcinolone, beclomethasone

How it works: Reduces inflammation in the nasal passages → less congestion, sneezing, runny nose.

INDICATIONS
- Allergic rhinitis — most effective drugs for it (first-line)
- Nasal polyps

MAIN SIDE EFFECTS
- Nasal burning, dryness, irritation
- Epistaxis (nosebleed)
- Sore throat, headache

ADVERSE EFFECTS
- Rare with nasal use: adrenal suppression, slowed growth in children at high doses
- Nasal septal perforation (rare)

CONTRAINDICATIONS & PRECAUTIONS
- Caution: nasal infection, ulcers, recent nasal surgery or trauma

SPECIAL WARNINGS
- Not for immediate relief — full effect may take 1–2 weeks
- Monitor growth in children on long-term use

CLIENT EDUCATION
- Clear (blow) your nose before spraying
- Use every day as prescribed for best effect
- Report frequent nosebleeds or signs of nasal infection
- No rebound congestion — safe for long-term use, unlike decongestant sprays

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Steroid nasal spray = slow but strong and safe long-term. Decongestant spray = fast but rebound after 3–5 days.

Ch 19 · Blood Pressure

RAAS (RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM) AGENTS

Captopril

Ch 19 · RAAS (renin-angiotensin-aldosterone system) agents

ACE (angiotensin-converting enzyme) inhibitor · also: lisinopril, enalapril, ramipril (all end in -pril)

How it works: Blocks conversion of angiotensin I to angiotensin II → vasodilation, less aldosterone (less sodium/water retention, more potassium kept).

INDICATIONS

- Hypertension
- Heart failure
- After MI (myocardial infarction) — limits remodeling, improves survival
- Diabetic and nondiabetic nephropathy (kidney protection)

MAIN SIDE EFFECTS

- Dry, persistent, nonproductive cough (bradykinin buildup)
- Rash and altered taste (captopril)
- Dizziness

ADVERSE EFFECTS

- First-dose hypotension — monitor BP (blood pressure) for several hours after first dose
- Hyperkalemia — monitor serum potassium
- ANGIOEDEMA (face, tongue, throat swelling) — stop drug, give epinephrine, protect airway
- Neutropenia (rare, captopril) — monitor WBC (white blood cell) count, report fever/sore throat

CONTRAINDICATIONS & PRECAUTIONS

- Pregnancy (fetal injury/death)
- History of angioedema
- Bilateral renal artery stenosis
- Caution: renal impairment, salt or volume depletion

INTERACTIONS

- Potassium-sparing diuretics, potassium supplements, salt substitutes → hyperkalemia; avoid, monitor K
- NSAIDs (nonsteroidal anti-inflammatory drugs) → reduced antihypertensive effect
- Lithium → lithium toxicity; monitor lithium levels
- Diuretics/other antihypertensives → additive first-dose hypotension

SPECIAL WARNINGS

- **BLACK BOX:** fetal toxicity — stop as soon as pregnancy is detected
- Check BP and potassium; watch renal function (BUN (blood urea nitrogen), creatinine)
- Cough is the most common reason clients stop — provider may switch to an ARB (angiotensin II receptor blocker)

CLIENT EDUCATION

- Take captopril 1 hour before meals — food reduces absorption
- Change positions slowly, especially after the first dose
- Report face, lip, tongue, or throat swelling right away — emergency
- Avoid salt substitutes and potassium supplements
- Tell your provider if you might be or plan to become pregnant

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: ACE inhibitor = cough + angioedema, both from bradykinin. -pril drugs: pregnancy is a hard no.

Losartan *(Cozaar)*

Ch 19 · RAAS (renin-angiotensin-aldosterone system) agents

ARB (angiotensin II receptor blocker) · also: valsartan, irbesartan, candesartan, olmesartan (all end in -sartan)

How it works: Blocks angiotensin II at its receptors → vasodilation and less aldosterone release; does not raise bradykinin.

INDICATIONS

- Hypertension
- Heart failure (alternative when ACE (angiotensin-converting enzyme) inhibitor not tolerated)
- Diabetic nephropathy
- Stroke prevention in hypertension with left ventricular hypertrophy

MAIN SIDE EFFECTS

- Dizziness
- Much less cough than ACE inhibitors

ADVERSE EFFECTS

- Angioedema — rare, but stop drug and protect airway
- Hyperkalemia — monitor potassium
- Hypotension — monitor BP (blood pressure)

CONTRAINDICATIONS & PRECAUTIONS

- Pregnancy (fetal injury/death)
- Caution: bilateral renal artery stenosis, renal impairment
- Caution: prior angioedema with an ACE inhibitor

INTERACTIONS

- Potassium supplements, potassium-sparing diuretics → hyperkalemia
- Other antihypertensives → additive hypotension
- Lithium → lithium toxicity

SPECIAL WARNINGS

- **BLACK BOX:** fetal toxicity — stop when pregnancy detected
- Monitor BP, potassium, and renal function

CLIENT EDUCATION

- Report swelling of face, lips, or tongue immediately
- Rise slowly to avoid dizziness
- Avoid salt substitutes containing potassium
- Notify provider if pregnant or planning pregnancy

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: ARB = ACE inhibitor benefits without the bradykinin cough. Still no pregnancy, still hyperkalemia.

Aliskiren *(Tekturna)*

Ch 19 · RAAS (renin-angiotensin-aldosterone system) agents

Direct renin inhibitor

How it works: Directly inhibits renin, the first step of the RAAS → less angiotensin II and aldosterone → lower BP (blood pressure).

INDICATIONS

- Hypertension

MAIN SIDE EFFECTS

- Diarrhea
- Cough (less than ACE (angiotensin-converting enzyme) inhibitors)
- Dizziness

ADVERSE EFFECTS

- Angioedema — stop drug, maintain airway
- Hyperkalemia — monitor potassium
- Hypotension

CONTRAINDICATIONS & PRECAUTIONS

- Pregnancy
- Do not combine with ACE inhibitors or ARBs (angiotensin II receptor blockers) in clients with diabetes
- Caution: renal impairment

INTERACTIONS

- High-fat meal → greatly reduced absorption; take consistently, avoid high-fat meals
- ACE inhibitors, ARBs, potassium supplements → hyperkalemia
- Furosemide → aliskiren lowers furosemide levels
- Ketoconazole, itraconazole, cyclosporine → raise aliskiren levels

SPECIAL WARNINGS

- **BLACK BOX:** fetal toxicity — stop when pregnancy detected
- Monitor BP, potassium, renal function

CLIENT EDUCATION

- Do not take with a high-fat meal
- Report swelling of face, lips, or tongue right away
- Report severe diarrhea

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Aliskiren blocks RAAS at the very first step (renin). High-fat meals tank absorption.

CALCIUM CHANNEL BLOCKERS

Nifedipine *(Procardia, Adalat)*

Ch 19 · Calcium channel blockers

CCB (calcium channel blocker) — dihydropyridine (acts on blood vessels) · also: amlodipine, felodipine, nicardipine (all end in -dipine)

How it works: Blocks calcium channels in arteriolar smooth muscle → arteriolar dilation → lower afterload and BP (blood pressure); little effect on heart.

INDICATIONS

- Hypertension
- Angina (including vasospastic/variant)

MAIN SIDE EFFECTS

- Headache, flushing, dizziness
- Peripheral edema (ankles/feet)
- Gingival hyperplasia

ADVERSE EFFECTS

- Reflex tachycardia — often given with a beta blocker to blunt it
- Hypotension — monitor BP and heart rate

CONTRAINDICATIONS & PRECAUTIONS

- Caution: immediate-release nifedipine is not used for hypertension (sudden BP drop)
- Caution: hypotension

INTERACTIONS

- Grapefruit juice → raises drug levels → toxicity; avoid
- Beta blockers → blunt reflex tachycardia (intended)
- Other antihypertensives → additive hypotension

SPECIAL WARNINGS

- Monitor BP, heart rate, and for edema
- Vessel-selective: does NOT slow the heart, so reflex tachycardia instead

CLIENT EDUCATION

- Swallow extended-release tablets whole — do not crush or chew
- Avoid grapefruit juice
- Report ankle swelling or fast heartbeat
- Good oral hygiene and dental visits for gum overgrowth

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Dihydropyridines (-dipine) = vessels only → reflex tachycardia + ankle edema, not bradycardia.

Verapamil *(Calan)*

Ch 19 · Calcium channel blockers

CCB (calcium channel blocker) — nondihydropyridine (acts on heart and vessels) · also: diltiazem

How it works: Blocks calcium channels in heart and arterioles → slower heart rate, less contractility, slower AV (atrioventricular) conduction, vasodilation.

INDICATIONS
- Hypertension
- Angina (stable and variant)
- Dysrhythmias: atrial fibrillation/flutter, SVT (supraventricular tachycardia)

MAIN SIDE EFFECTS
- Constipation (most common with verapamil)
- Dizziness, headache, flushing
- Peripheral edema

ADVERSE EFFECTS
- Bradycardia — check pulse before dose
- Heart block (AV block) — monitor ECG (electrocardiogram)
- Hypotension
- Worsening heart failure (reduced contractility)

CONTRAINDICATIONS & PRECAUTIONS
- Severe hypotension
- Sick sinus syndrome, 2nd/3rd-degree AV block (without pacemaker)
- Caution: heart failure

INTERACTIONS
- Digoxin → verapamil raises digoxin level + additive AV slowing; monitor digoxin level, HR (heart rate)
- Beta blockers → additive bradycardia, heart block, HF (heart failure); monitor closely
- Grapefruit juice → raises drug levels; avoid

SPECIAL WARNINGS
- Monitor heart rate, BP (blood pressure), ECG; report HR below 60 per provider order
- Watch for HF signs: edema, weight gain, dyspnea

CLIENT EDUCATION
- Increase fiber and fluids to prevent constipation
- Swallow extended-release tablets whole
- Avoid grapefruit juice
- Report slow pulse, dizziness, fainting, swelling, or shortness of breath

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Verapamil/diltiazem hit the HEART: bradycardia, heart block, constipation, and they raise digoxin.

SYMPATHOLYTICS

Prazosin *(Minipress)*

Ch 19 · Sympatholytics

Alpha-1 adrenergic blocker · also: doxazosin, terazosin (all end in -zosin)

How it works: Blocks alpha-1 receptors on arterioles and veins → vasodilation → lower BP (blood pressure).

INDICATIONS
- Hypertension
- BPH (benign prostatic hyperplasia) — doxazosin, terazosin

MAIN SIDE EFFECTS
- Dizziness, drowsiness, headache
- Nasal congestion
- Inhibited ejaculation

ADVERSE EFFECTS
- First-dose orthostatic hypotension — fainting possible; start low dose, at bedtime
- Reflex tachycardia
- Sodium/water retention, edema

CONTRAINDICATIONS & PRECAUTIONS
- Known hypersensitivity
- Caution: concurrent PDE-5 (phosphodiesterase-5) inhibitors like sildenafil

INTERACTIONS
- Other antihypertensives, PDE-5 inhibitors (sildenafil) → additive hypotension

SPECIAL WARNINGS
- Monitor BP lying and standing, especially after first dose
- Fall precautions at start of therapy

CLIENT EDUCATION
- Take the first dose at bedtime
- Rise slowly; sit or lie down if dizzy or lightheaded
- Avoid driving or hazardous tasks for 12–24 hours after first dose

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: -zosin = first-dose syncope → bedtime first dose, rise slowly.

Clonidine *(Catapres)*

Ch 19 · Sympatholytics

Centrally acting alpha-2 adrenergic agonist · also: methyldopa, guanfacine

How it works: Stimulates alpha-2 receptors in the brainstem → less sympathetic outflow → lower heart rate and vasodilation.

INDICATIONS
- Hypertension
- ADHD (attention-deficit/hyperactivity disorder)
- Opioid withdrawal symptoms
- Severe cancer pain (epidural)

MAIN SIDE EFFECTS
- Dry mouth
- Drowsiness, sedation
- Constipation

ADVERSE EFFECTS
- Rebound hypertension if stopped abruptly — taper over 2–4 days
- Bradycardia, hypotension

CONTRAINDICATIONS & PRECAUTIONS
- Caution: pregnancy
- Caution: heart block, severe coronary disease

INTERACTIONS
- Alcohol, CNS (central nervous system) depressants → additive sedation
- Other antihypertensives → additive hypotension
- Beta blockers → worse rebound if clonidine stopped; stop beta blocker first

SPECIAL WARNINGS
- Never stop abruptly — rebound hypertension can be severe
- Monitor BP (blood pressure) and heart rate

CLIENT EDUCATION
- Do not stop suddenly — taper with provider
- Sip water, chew sugarless gum for dry mouth
- Avoid driving until you know how drowsy it makes you
- Patch: change every 7 days, rotate sites, remove before MRI (magnetic resonance imaging)

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Clonidine: dry mouth + drowsy; abrupt stop = rebound hypertension. Patch is weekly.

Metoprolol *(Lopressor, Toprol-XL)*

Ch 19 · Sympatholytics

Beta blocker — cardioselective (beta-1) · also: atenolol, bisoprolol, esmolol (all end in -olol)

How it works: Blocks beta-1 receptors in heart → lower heart rate, contractility, and renin release → lower BP (blood pressure) and O2 demand.

INDICATIONS

- Hypertension
- Angina
- After MI (myocardial infarction)
- Heart failure (metoprolol succinate)
- Tachydysrhythmias

MAIN SIDE EFFECTS

- Fatigue, dizziness
- Sexual dysfunction
- Depression

ADVERSE EFFECTS

- Bradycardia, AV (atrioventricular) block — hold if apical pulse <60 and notify provider
- Hypotension
- Worsening heart failure — watch edema, weight gain, dyspnea
- Rebound angina/MI/dysrhythmias if stopped abruptly

CONTRAINDICATIONS & PRECAUTIONS

- Sinus bradycardia, 2nd/3rd-degree AV block
- Cardiogenic shock, decompensated heart failure
- Caution: diabetes (masks hypoglycemia)
- Caution: asthma/COPD (chronic obstructive pulmonary disease) — safer than nonselective

INTERACTIONS

- Verapamil, diltiazem → additive bradycardia, heart block
- Insulin, oral antidiabetics → masked hypoglycemia; monitor glucose
- Other antihypertensives → additive hypotension

SPECIAL WARNINGS

- **BLACK BOX:** do not stop abruptly — taper over 1–2 weeks (risk of angina, MI)
- Check apical pulse 1 full minute; hold if <60/min
- Masks tachycardia and tremor of hypoglycemia; sweating still occurs

CLIENT EDUCATION

- Take your pulse daily; call provider if below 60
- Never stop suddenly
- If diabetic, check glucose often — warning signs may be hidden
- Rise slowly to prevent dizziness

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Cardioselective (metoprolol, atenolol) = beta-1: you have 1 heart. Beta-2: you have 2 lungs.

Propranolol *(Inderal)*

Ch 19 · Sympatholytics

Beta blocker — nonselective (beta-1 and beta-2) · also: nadolol, timolol (carvedilol, labetalol also block alpha-1)

How it works: Blocks beta-1 (heart) and beta-2 (lungs, vessels, liver) receptors → lower heart rate and BP (blood pressure), but also bronchoconstriction.

INDICATIONS

- Hypertension
- Angina
- Dysrhythmias
- Migraine prevention
- Performance anxiety, essential tremor

MAIN SIDE EFFECTS

- Fatigue, dizziness
- Sexual dysfunction
- Depression, sleep disturbance

ADVERSE EFFECTS

- Bronchospasm (beta-2 block) — avoid in asthma
- Bradycardia, AV (atrioventricular) block — hold if apical pulse <60
- Masked and prolonged hypoglycemia (blocks liver glucose release)
- Rebound cardiac excitation if stopped abruptly

CONTRAINDICATIONS & PRECAUTIONS

- Asthma, bronchospasm, severe COPD (chronic obstructive pulmonary disease)
- Sinus bradycardia, 2nd/3rd-degree AV block
- Cardiogenic shock, decompensated heart failure
- Caution: diabetes

INTERACTIONS

- Verapamil, diltiazem → additive bradycardia, heart block
- Insulin → masked, prolonged hypoglycemia; monitor glucose
- Other antihypertensives → additive hypotension

SPECIAL WARNINGS

- **BLACK BOX:** do not stop abruptly — taper over 1–2 weeks
- Assess lung sounds and breathing; check apical pulse 1 full minute

CLIENT EDUCATION

- Report wheezing or shortness of breath
- Take your pulse daily; report below 60
- Never stop suddenly
- Diabetics: monitor glucose closely

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Nonselective = beta-1 AND beta-2: 1 heart, 2 lungs → bronchospasm. No propranolol in asthma.

DIRECT-ACTING VASODILATORS

Hydralazine

Ch 19 · Direct-acting vasodilators

Direct-acting arteriolar vasodilator · also: minoxidil

How it works: Relaxes arteriolar smooth muscle directly → lower afterload and BP (blood pressure).

INDICATIONS
- Hypertension (oral)
- Hypertensive crisis, including severe hypertension in pregnancy (IV (intravenous))
- Heart failure — with isosorbide dinitrate

MAIN SIDE EFFECTS
- Headache, dizziness
- Flushing
- Nausea

ADVERSE EFFECTS
- Reflex tachycardia — given with a beta blocker
- Sodium and water retention, edema — given with a diuretic
- SLE (systemic lupus erythematosus)-like syndrome: joint/muscle pain, fever — stop drug, report

CONTRAINDICATIONS & PRECAUTIONS
- Coronary artery disease (reflex tachycardia can trigger angina)
- Caution: SLE

INTERACTIONS
- Other antihypertensives → additive hypotension
- Beta blockers, diuretics → offset reflex tachycardia and fluid retention (intended)

SPECIAL WARNINGS
- Monitor BP and heart rate
- Monitor daily weight and edema
- Watch for lupus-like symptoms with long-term use

CLIENT EDUCATION
- Report joint pain, fever, muscle aches, or rash
- Weigh daily; report weight gain or swelling
- Rise slowly

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Hydralazine triad: reflex tachycardia, fluid retention, lupus-like syndrome.

Nitroprusside *(Nitropress)*

Ch 19 · Direct-acting vasodilators

Direct-acting vasodilator (arterial and venous)

How it works: Releases nitric oxide → dilates arteries and veins → immediate drop in BP (blood pressure), preload, and afterload.

INDICATIONS
- Hypertensive crisis/emergency (continuous IV (intravenous) infusion)
- Acute decompensated heart failure

MAIN SIDE EFFECTS
- Headache
- Nausea, restlessness, sweating (from rapid BP drop)

ADVERSE EFFECTS
- Profound hypotension — continuous BP monitoring, titrate carefully
- Cyanide toxicity: metabolic acidosis, low O2 sat, bradycardia, confusion, seizures
- Thiocyanate toxicity (risk with renal impairment, >3 days): tinnitus, blurred vision, delirium, psychosis

CONTRAINDICATIONS & PRECAUTIONS
- Caution: renal or liver impairment (thiocyanate/cyanide buildup)

INTERACTIONS
- Other antihypertensives → additive hypotension

SPECIAL WARNINGS
- Protect IV bag from light; faint brown tint OK, discard if strongly colored
- Continuous BP monitoring (often arterial line), infusion pump only
- Monitor thiocyanate levels with prolonged infusion; limit to shortest time

CLIENT EDUCATION
- Report headache, ringing in ears, blurred vision, or confusion

ANTIDOTE
- For cyanide toxicity: stop infusion; sodium thiosulfate or hydroxocobalamin

Remember: Nitroprusside: works in seconds, protect from light, watch cyanide/thiocyanate.

Ch 20 · Cardiac Glycosides & Heart Failure

CARDIAC GLYCOSIDES

Digoxin *(Lanoxin)*

Ch 20 · Cardiac glycosides

Cardiac glycoside (positive inotrope, negative chronotrope)

How it works: Inhibits sodium-potassium ATPase pump → more intracellular calcium → stronger contraction; slows SA (sinoatrial) and AV (atrioventricular) conduction.

INDICATIONS

- Heart failure (symptom control; does not improve survival)
- Atrial fibrillation/flutter — ventricular rate control

MAIN SIDE EFFECTS

- Anorexia, nausea, vomiting (often first sign of toxicity)
- Fatigue, weakness

ADVERSE EFFECTS

- Toxicity — GI (gastrointestinal): anorexia, nausea, vomiting, abdominal pain
- Toxicity — visual: blurred vision, yellow-green halos around lights
- Toxicity — cardiac: bradycardia, any new dysrhythmia (most dangerous)
- If toxic: hold dose, check digoxin and potassium levels, notify provider

CONTRAINDICATIONS & PRECAUTIONS

- Ventricular fibrillation
- Caution: hypokalemia, hypomagnesemia, hypercalcemia
- Caution: renal impairment, older adults
- Caution: heart block, bradycardia

INTERACTIONS

- Loop/thiazide diuretics → hypokalemia → toxicity; monitor K
- Verapamil, amiodarone, quinidine → raise digoxin level; monitor level
- ACE (angiotensin-converting enzyme) inhibitors, ARBs, spironolactone → hyperkalemia → reduced digoxin effect
- Antacids → reduced absorption; separate doses
- Dobutamine, other sympathomimetics → higher dysrhythmia risk

SPECIAL WARNINGS

- Therapeutic level: 0.5–2 ng/mL (narrow therapeutic index)
- Count apical pulse 1 full minute; hold if <60 adult, <70 child, <90 infant
- Hypokalemia (K <3.5 mEq/L) is the biggest toxicity risk
- Draw level at least 6–8 hours after a dose (or right before next dose)

CLIENT EDUCATION

- Take your pulse before each dose; call provider if below 60
- Take at the same time daily; never double a missed dose
- Report loss of appetite, nausea, vomiting, or visual changes
- Eat potassium-rich foods if also taking a potassium-wasting diuretic
- Separate from antacids by about 2 hours

ANTIDOTE

- Digoxin immune fab (Digibind, DigiFab)

Remember: Low potassium = digoxin toxicity at a normal level. Yellow-green halos are the classic exam clue.

HEART FAILURE — RAAS AGENTS

ACE inhibitors and ARBs in heart failure

Ch 20 · Heart failure — RAAS agents

ACE (angiotensin-converting enzyme) inhibitors / ARBs (angiotensin II receptor blockers) · also: captopril, lisinopril, enalapril; losartan, valsartan, candesartan

How it works: Block angiotensin II formation or action → lower afterload and preload, less aldosterone, less cardiac remodeling.

INDICATIONS

- Heart failure — first-line; improve survival and slow progression
- After MI (myocardial infarction) to limit ventricular remodeling

MAIN SIDE EFFECTS

- Dry cough (ACE inhibitors only)
- Dizziness

ADVERSE EFFECTS

- First-dose hypotension — monitor BP (blood pressure)
- Hyperkalemia — monitor potassium
- Angioedema — stop drug, maintain airway
- Worsening renal function — monitor creatinine

CONTRAINDICATIONS & PRECAUTIONS

- Pregnancy
- History of angioedema
- Bilateral renal artery stenosis

INTERACTIONS

- Spironolactone, potassium supplements → hyperkalemia; monitor K
- Diuretics → additive hypotension
- NSAIDs (nonsteroidal anti-inflammatory drugs) → reduced effect, renal injury

SPECIAL WARNINGS

- **BLACK BOX:** fetal toxicity
- Start at low dose and titrate up; monitor BP, K, renal function

CLIENT EDUCATION

- Rise slowly
- Report swelling of face, lips, or tongue immediately
- Avoid potassium supplements and salt substitutes unless prescribed
- Weigh daily; report gain over 2 lb in a day or 5 lb in a week

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: In heart failure, ACE inhibitors/ARBs are survival drugs; digoxin only treats symptoms.

Sacubitril/valsartan *(Entresto)*

Ch 20 · Heart failure — RAAS agents

ARNI (angiotensin receptor-neprilysin inhibitor)

How it works: Sacubitril blocks neprilysin (raises natriuretic peptides → vasodilation, sodium excretion); valsartan blocks angiotensin II receptors.

INDICATIONS

- Chronic heart failure with reduced ejection fraction — lowers hospitalization and death

MAIN SIDE EFFECTS

- Dizziness
- Cough

ADVERSE EFFECTS

- Hypotension — monitor BP (blood pressure)
- Hyperkalemia — monitor potassium
- Angioedema — stop drug, maintain airway
- Decreased renal function — monitor creatinine

CONTRAINDICATIONS & PRECAUTIONS

- Pregnancy
- History of angioedema with ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin II receptor blocker)
- Use with an ACE inhibitor or within 36 hours of last ACE inhibitor dose
- With aliskiren in clients with diabetes

INTERACTIONS

- ACE inhibitors → angioedema; allow 36-hour washout
- Potassium supplements, K-sparing diuretics → hyperkalemia
- Lithium → lithium toxicity
- NSAIDs (nonsteroidal anti-inflammatory drugs) → renal injury

SPECIAL WARNINGS

- **BLACK BOX:** fetal toxicity
- 36-HOUR washout when switching from an ACE inhibitor
- Replaces the ACE inhibitor or ARB — never given with one

CLIENT EDUCATION

- Do not take your old ACE inhibitor once this starts
- Report face, lip, tongue, or throat swelling at once
- Rise slowly; report dizziness or fainting
- Notify provider if pregnant

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Entresto + ACE inhibitor overlap = angioedema. Stop the -pril, wait 36 hours.

HEART FAILURE — BETA BLOCKERS

Carvedilol *(Coreg)*

Ch 20 · Heart failure — beta blockers

Beta blocker (nonselective beta + alpha-1 blocker) · also: metoprolol succinate (extended-release), bisoprolol

How it works: Blocks chronic sympathetic overstimulation of the failing heart → slower rate, less remodeling; alpha-1 block adds vasodilation.

INDICATIONS

- Chronic heart failure — improves survival
- Hypertension
- After MI (myocardial infarction)

MAIN SIDE EFFECTS

- Dizziness, fatigue
- Weight gain

ADVERSE EFFECTS

- Heart failure symptoms may worsen at first — monitor weight, edema, dyspnea
- Bradycardia, AV (atrioventricular) block — hold if apical pulse <60
- Hypotension, orthostatic dizziness
- Bronchospasm (carvedilol nonselective)

CONTRAINDICATIONS & PRECAUTIONS

- Decompensated (acute) heart failure
- Sinus bradycardia, 2nd/3rd-degree AV block
- Asthma (carvedilol)
- Caution: diabetes (masks hypoglycemia)

INTERACTIONS

- Verapamil, diltiazem → additive bradycardia, heart block
- Digoxin → additive bradycardia
- Insulin → masked hypoglycemia

SPECIAL WARNINGS

- START LOW, GO SLOW — dose doubled about every 2 weeks as tolerated
- Do not stop abruptly
- Check apical pulse and BP (blood pressure) before each dose

CLIENT EDUCATION

- You may feel worse for the first few weeks — keep taking it
- Weigh daily; report weight gain or swelling
- Take carvedilol with food to reduce dizziness
- Never stop suddenly

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Beta blockers in heart failure: worse before better. Only carvedilol, metoprolol succinate, bisoprolol proven.

HEART FAILURE — DIURETICS

Furosemide *(Lasix)*

Ch 20 · Heart failure — diuretics

Loop diuretic · also: bumetanide, torsemide

How it works: Blocks sodium and chloride reabsorption in loop of Henle → large diuresis → lower preload and pulmonary congestion.

INDICATIONS

- Heart failure — pulmonary and peripheral edema
- Pulmonary edema
- Edema from liver or kidney disease
- Hypertension (when thiazide not enough)

MAIN SIDE EFFECTS

- Frequent urination
- Dizziness

ADVERSE EFFECTS

- Hypokalemia — monitor K, give supplements as ordered
- Dehydration, hypotension — monitor I&O (intake and output), BP (blood pressure)
- Hyponatremia, hypomagnesemia
- Ototoxicity (tinnitus, hearing loss) — IV push no faster than 20 mg/min
- Hyperglycemia, hyperuricemia (gout)

CONTRAINDICATIONS & PRECAUTIONS

- Anuria
- Caution: hypokalemia, dehydration
- Caution: diabetes, gout

INTERACTIONS

- Digoxin → hypokalemia → digoxin toxicity; monitor K
- Aminoglycosides (gentamicin) → additive ototoxicity
- Lithium → lithium toxicity
- NSAIDs (nonsteroidal anti-inflammatory drugs) → reduced diuretic effect
- Other antihypertensives → additive hypotension

SPECIAL WARNINGS

- **BLACK BOX:** potent diuretic — excess can cause severe fluid/electrolyte loss
- Daily weights, I&O, potassium levels
- Give in the morning to avoid nighttime urination

CLIENT EDUCATION

- Weigh daily, same time and scale; report gain over 2 lb/day or 5 lb/week
- Eat potassium-rich foods (bananas, potatoes, oranges, spinach)
- Report muscle cramps, weakness, or ringing in the ears
- Rise slowly to prevent dizziness

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Furosemide + digoxin = the classic hypokalemia → digoxin toxicity setup.

Spironolactone *(Aldactone)*

Ch 20 · Heart failure — diuretics

Potassium-sparing diuretic (aldosterone antagonist) · also: eplerenone

How it works: Blocks aldosterone in distal tubule → excretes sodium/water, retains potassium; reduces cardiac remodeling.

INDICATIONS

- Heart failure — improves survival
- Hypertension
- Primary hyperaldosteronism
- Edema (often paired with a loop or thiazide diuretic)

MAIN SIDE EFFECTS

- Gynecomastia (men)
- Menstrual irregularities, deepening voice, hirsutism
- Erectile dysfunction

ADVERSE EFFECTS

- Hyperkalemia — monitor potassium; watch for dysrhythmias
- Dehydration, hyponatremia

CONTRAINDICATIONS & PRECAUTIONS

- Hyperkalemia
- Severe renal impairment
- Caution: use with ACE (angiotensin-converting enzyme) inhibitors/ARBs (angiotensin II receptor blockers)

INTERACTIONS

- ACE inhibitors, ARBs, potassium supplements, salt substitutes → hyperkalemia; monitor K
- Loop/thiazide diuretics → offsets potassium loss (intended)

SPECIAL WARNINGS

- Monitor serum potassium closely, especially with ACE inhibitors/ARBs or renal impairment
- Daily weights, I&O (intake and output)

CLIENT EDUCATION

- Avoid potassium supplements, salt substitutes, and excess high-potassium foods
- Report muscle weakness, irregular heartbeat
- Report breast enlargement or menstrual changes

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Spironolactone SPARES potassium → hyperkalemia; blocks androgen effects → gynecomastia.

INOTROPIC AGENTS

Dobutamine

Ch 20 · Inotropic agents

Beta-1 adrenergic agonist (catecholamine inotrope) · also: dopamine

How it works: Stimulates beta-1 receptors in heart → stronger contraction → higher cardiac output.

INDICATIONS
- Acute decompensated heart failure — short-term IV (intravenous) support
- Cardiogenic shock / low cardiac output

MAIN SIDE EFFECTS
- Headache, nausea
- Palpitations

ADVERSE EFFECTS
- Tachycardia, tachydysrhythmias — continuous ECG (electrocardiogram) monitoring
- Hypertension or hypotension — monitor BP (blood pressure) closely
- Angina (increased O2 demand)

CONTRAINDICATIONS & PRECAUTIONS
- Idiopathic hypertrophic subaortic stenosis
- Caution: atrial fibrillation (speeds AV (atrioventricular) conduction)

INTERACTIONS
- Beta blockers → reduce dobutamine effect
- Digoxin → higher dysrhythmia risk

SPECIAL WARNINGS
- Continuous IV infusion by pump only
- Continuous ECG, BP, and urine output monitoring

CLIENT EDUCATION
- Report chest pain, palpitations, or shortness of breath

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Dobutamine = beta-1 = stronger beat; the risk is a racing, irregular heart.

Milrinone

Ch 20 · Inotropic agents

Phosphodiesterase-3 inhibitor (inodilator)

How it works: Inhibits phosphodiesterase → more cAMP (cyclic adenosine monophosphate) → stronger contraction plus vasodilation (lower afterload and preload).

INDICATIONS
- Acute decompensated heart failure — short-term IV (intravenous) therapy

MAIN SIDE EFFECTS
- Headache
- Hypokalemia

ADVERSE EFFECTS
- Ventricular dysrhythmias — continuous ECG (electrocardiogram)
- Hypotension — monitor BP (blood pressure)

CONTRAINDICATIONS & PRECAUTIONS
- Caution: renal impairment (dose reduce)
- Caution: severe aortic or pulmonic valve disease

INTERACTIONS
- Other antihypertensives, vasodilators → additive hypotension

SPECIAL WARNINGS
- IV infusion by pump; short-term use only
- Monitor ECG, BP, fluid status, potassium, and renal function

CLIENT EDUCATION
- Report palpitations, dizziness, or chest pain

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Milrinone = 'inodilator': pumps harder AND dilates, so watch for hypotension.

NATRIURETIC PEPTIDES

Nesiritide *(Natrecor)*

Ch 20 · Natriuretic peptides

Recombinant BNP (B-type natriuretic peptide)

How it works: Acts like natural BNP → dilates arteries and veins, promotes sodium and water excretion → lower preload and afterload.

INDICATIONS
- Acutely decompensated heart failure with dyspnea at rest (IV (intravenous))

MAIN SIDE EFFECTS
- Headache
- Nausea

ADVERSE EFFECTS
- Hypotension — the major adverse effect; monitor BP (blood pressure) continuously
- Possible decline in renal function — monitor creatinine

CONTRAINDICATIONS & PRECAUTIONS
- Cardiogenic shock
- Low baseline systolic BP

INTERACTIONS
- Other antihypertensives, ACE (angiotensin-converting enzyme) inhibitors, nitrates → additive hypotension

SPECIAL WARNINGS
- IV bolus then continuous infusion; monitor BP, heart rate, urine output

CLIENT EDUCATION
- Report dizziness or lightheadedness

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Nesiritide is BNP as a drug — so expect vasodilation and hypotension.

HEART FAILURE — HEART-RATE REDUCTION

Ivabradine *(Corlanor)*

Ch 20 · Heart failure — heart-rate reduction

HCN (hyperpolarization-activated cyclic nucleotide-gated) channel blocker

How it works: Blocks the 'funny' current in the SA (sinoatrial) node → slows heart rate without lowering contractility.

INDICATIONS
- Stable chronic heart failure with reduced ejection fraction, sinus rhythm, resting HR (heart rate) ≥70 on max beta blocker

MAIN SIDE EFFECTS
- Luminous phenomena (phosphenes) — brief bright spots in vision, usually fade
- Hypertension

ADVERSE EFFECTS
- Bradycardia — monitor heart rate
- Atrial fibrillation — monitor rhythm
- Fetal toxicity

CONTRAINDICATIONS & PRECAUTIONS
- Pregnancy
- Acute decompensated heart failure
- BP (blood pressure) below 90/50
- Sick sinus syndrome, SA block, 3rd-degree AV (atrioventricular) block without pacemaker
- Resting HR below 60 before treatment; severe liver impairment

INTERACTIONS
- Strong CYP3A4 inhibitors (ketoconazole, clarithromycin) → contraindicated
- Grapefruit juice → raises levels; avoid
- Beta blockers, digoxin, verapamil, diltiazem → additive bradycardia

SPECIAL WARNINGS
- Women of childbearing age need effective contraception
- Monitor HR and rhythm; report HR below 50 or symptoms

CLIENT EDUCATION
- Take twice daily with meals
- Brief flashes of bright light in vision may occur — use caution driving in changing light
- Use reliable birth control; report pregnancy
- Report dizziness, fatigue, or irregular/racing heartbeat
- Avoid grapefruit juice

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Ivabradine slows only the SA node: bradycardia + 'seeing bright spots' (luminous phenomena).

HEART FAILURE — SGLT2 INHIBITORS

Dapagliflozin *(Farxiga)*

Ch 20 · Heart failure — SGLT2 inhibitors

SGLT2 (sodium-glucose cotransporter 2) inhibitor · also: empagliflozin (Jardiance), canagliflozin

How it works: Blocks glucose and sodium reabsorption in proximal tubule → glucose and sodium lost in urine → mild diuresis.

INDICATIONS

- Heart failure (reduced or preserved ejection fraction) — lowers HF hospitalization and death
- Type 2 diabetes mellitus
- Chronic kidney disease

MAIN SIDE EFFECTS

- Genital fungal (yeast) infections
- UTIs (urinary tract infections)
- Increased urination

ADVERSE EFFECTS

- Volume depletion, hypotension — monitor BP (blood pressure), orthostatic changes
- Ketoacidosis, even with normal glucose — check ketones if ill
- Fournier gangrene (rare perineal infection) — report pain, swelling
- Hypoglycemia with insulin or sulfonylureas

CONTRAINDICATIONS & PRECAUTIONS

- Dialysis
- Caution: type 1 diabetes (not indicated; ketoacidosis risk)
- Caution: dehydration, older adults

INTERACTIONS

- Loop diuretics, antihypertensives → additive volume depletion, hypotension
- Insulin, sulfonylureas → hypoglycemia; dose may need lowering

SPECIAL WARNINGS

- Hold about 3 days before planned surgery (ketoacidosis risk)
- Monitor renal function, volume status

CLIENT EDUCATION

- Keep genital area clean and dry; report itching, discharge, or pain
- Drink fluids; report dizziness or lightheadedness
- Report nausea, vomiting, abdominal pain, unusual fatigue (possible ketoacidosis)
- Glucose in urine is expected — urine tests will read positive

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: SGLT2 = sugar out in the urine → yeast infections, dehydration, ketoacidosis.

Ch 21 · Angina

ORGANIC NITRATES

Nitroglycerin *(Nitrostat)*

Ch 21 · Organic nitrates

Organic nitrate (vasodilator) · also: isosorbide mononitrate, isosorbide dinitrate

How it works: Converted to nitric oxide → mainly venous dilation → lower preload and O2 demand; also dilates coronary arteries.

INDICATIONS

- Acute angina attack (sublingual tablet or translingual spray)
- Prevention of angina before exertion
- Long-term prophylaxis (transdermal patch, ointment)
- Unstable angina, acute MI (myocardial infarction), acute heart failure (IV (intravenous))

MAIN SIDE EFFECTS

- Headache (most common) — treat with acetaminophen; usually fades
- Facial flushing
- Dizziness

ADVERSE EFFECTS

- Orthostatic hypotension — sit or lie down for dose; rise slowly
- Reflex tachycardia — may add a beta blocker
- Tolerance with continuous use — needs nitrate-free period

CONTRAINDICATIONS & PRECAUTIONS

- Use of a PDE-5 (phosphodiesterase-5) inhibitor: sildenafil, tadalafil, vardenafil
- Hypotension
- Caution: head injury (can raise intracranial pressure)

INTERACTIONS

- PDE-5 inhibitors (sildenafil) → life-threatening hypotension; never combine
- Alcohol, beta blockers, CCBs (calcium channel blockers), diuretics → additive hypotension

SPECIAL WARNINGS

- Sublingual: 1 tablet every 5 min, max 3 doses
- Call 911 if pain not relieved 5 min after FIRST dose
- Transdermal: 10–12 hour nitrate-free period daily (usually overnight) to prevent tolerance
- IV: glass bottle and non-PVC (polyvinyl chloride) tubing; continuous BP (blood pressure) monitoring
- Wear gloves applying patch/ointment; remove old patch, rotate sites

CLIENT EDUCATION

- Sit down, place tablet under tongue, let it dissolve — don't swallow
- Store in the original dark glass bottle, tightly capped, away from heat/moisture
- Carry tablets at all times; replace per expiration date
- Tell every provider you take nitrates — never use erectile dysfunction drugs
- Take acetaminophen for headache; don't skip doses because of it

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: One tablet, wait 5 minutes, no relief → call 911 before taking the next one.

Isosorbide mononitrate *(Imdur)*

Ch 21 · Organic nitrates

Organic nitrate — long-acting oral · also: isosorbide dinitrate (Isordil)

How it works: Converted to nitric oxide → venous dilation → lower preload and myocardial O2 demand over many hours.

INDICATIONS

- Prevention (prophylaxis) of stable angina — not for acute attacks
- Heart failure — isosorbide dinitrate with hydralazine

MAIN SIDE EFFECTS

- Headache
- Dizziness, flushing

ADVERSE EFFECTS

- Orthostatic hypotension — rise slowly
- Reflex tachycardia
- Tolerance — dose so there is a daily nitrate-free period

CONTRAINDICATIONS & PRECAUTIONS

- Use of PDE-5 (phosphodiesterase-5) inhibitors: sildenafil, tadalafil, vardenafil
- Hypotension

INTERACTIONS

- PDE-5 inhibitors → life-threatening hypotension
- Alcohol, other antihypertensives → additive hypotension

SPECIAL WARNINGS

- Long-acting oral nitrates prevent attacks; client still needs sublingual nitroglycerin for acute pain
- Do not stop abruptly — can trigger angina

CLIENT EDUCATION

- Take on the exact schedule prescribed (extended-release usually once each morning)
- Swallow extended-release tablets whole
- Keep sublingual nitroglycerin on hand for chest pain
- Avoid alcohol and erectile dysfunction drugs

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Isosorbide = prevention; sublingual nitroglycerin = rescue. Same nitrate side effects and PDE-5 ban.

ANTIANGINAL — BETA BLOCKERS

Metoprolol (for angina) *(Lopressor, Toprol-XL)*

Ch 21 · Antianginal — beta blockers

Beta blocker (beta-1 cardioselective) · also: atenolol, propranolol

How it works: Blocks beta-1 → slower heart rate and less contractility → lower O2 demand; longer diastole improves coronary filling.

INDICATIONS

- Chronic stable (exertional) angina — first-line prevention
- After MI (myocardial infarction) — reduces mortality

MAIN SIDE EFFECTS

- Fatigue, dizziness
- Sexual dysfunction

ADVERSE EFFECTS

- Bradycardia, AV (atrioventricular) block — hold if apical pulse <60
- Hypotension
- Rebound angina or MI if stopped abruptly

CONTRAINDICATIONS & PRECAUTIONS

- Not used alone for variant (vasospastic/Prinzmetal) angina
- Sinus bradycardia, 2nd/3rd-degree AV block
- Caution: asthma, diabetes

INTERACTIONS

- Verapamil, diltiazem → additive bradycardia, heart block
- Nitrates → blunts nitrate reflex tachycardia (intended)

SPECIAL WARNINGS

- **BLACK BOX:** do not stop abruptly — taper over 1–2 weeks
- Check apical pulse and BP (blood pressure) before dosing

CLIENT EDUCATION

- Take every day to prevent attacks — it does not treat an acute attack
- Never stop suddenly
- Take pulse daily; report below 60

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Beta blockers prevent exertional angina (lower demand) but don't help vasospasm.

ANTIANGINAL — CALCIUM CHANNEL BLOCKERS

Verapamil, diltiazem, amlodipine (for angina)

Ch 21 · Antianginal — calcium channel blockers

CCBs (calcium channel blockers) · also: nifedipine (extended-release only)

How it works: Dilate arterioles (lower afterload) and relax coronary arteries; verapamil/diltiazem also slow heart rate → lower O2 demand.

INDICATIONS
- Variant (vasospastic/Prinzmetal) angina — relieve coronary spasm (drug of choice)
- Chronic stable angina — lower O2 demand, alternative or add-on to beta blockers

MAIN SIDE EFFECTS
- Constipation (verapamil)
- Peripheral edema, flushing, headache (amlodipine, nifedipine)

ADVERSE EFFECTS
- Bradycardia, AV (atrioventricular) block (verapamil, diltiazem) — check pulse
- Reflex tachycardia (nifedipine, amlodipine)
- Hypotension

CONTRAINDICATIONS & PRECAUTIONS
- Verapamil/diltiazem: AV block, sick sinus syndrome, caution in heart failure
- Caution: immediate-release nifedipine avoided (sudden BP (blood pressure) drop)

INTERACTIONS
- Grapefruit juice → raises CCB levels; avoid
- Beta blockers + verapamil/diltiazem → bradycardia, heart block
- Verapamil → raises digoxin level

SPECIAL WARNINGS
- Monitor heart rate, BP, edema
- Long-acting forms preferred for angina

CLIENT EDUCATION
- Take daily for prevention, not for an acute attack
- Avoid grapefruit juice
- Swallow extended-release forms whole
- Report slow pulse, swelling, or dizziness

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Variant angina = spasm → calcium channel blockers (beta blockers can't relax spasm).

ANTIANGINAL — OTHER

Ranolazine *(Ranexa)*

Ch 21 · Antianginal — other

Antianginal (late sodium current inhibitor)

How it works: Inhibits late sodium current in heart cells → less calcium overload → less O2 demand; does not change heart rate or BP (blood pressure).

INDICATIONS
- Chronic stable angina — usually added when other antianginals aren't enough

MAIN SIDE EFFECTS
- Dizziness, headache
- Constipation, nausea

ADVERSE EFFECTS
- QT interval prolongation — baseline and periodic ECG (electrocardiogram)
- Renal failure in severe renal impairment — monitor creatinine

CONTRAINDICATIONS & PRECAUTIONS
- Liver cirrhosis
- Strong CYP3A4 inhibitors: ketoconazole, clarithromycin, ritonavir
- CYP3A4 inducers: rifampin, St. John's wort
- Caution: existing QT prolongation or other QT-prolonging drugs

INTERACTIONS
- Ketoconazole, grapefruit juice → raise ranolazine level → QT prolongation; avoid
- Diltiazem, verapamil → raise ranolazine level; dose limited
- Ranolazine → raises digoxin and simvastatin levels

SPECIAL WARNINGS
- Monitor ECG for QT prolongation
- Not for acute angina attacks

CLIENT EDUCATION
- Swallow extended-release tablets whole
- Avoid grapefruit juice
- Report fainting, palpitations, or dizziness
- Keep nitroglycerin for acute chest pain

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Ranolazine = QT prolongation + CYP3A4 (no ketoconazole, no grapefruit, no cirrhosis).

ANTIPLATELETS

Aspirin (low-dose)

Ch 21 · Antiplatelets

Antiplatelet (irreversible COX (cyclooxygenase) inhibitor) · also: clopidogrel

How it works: Irreversibly blocks COX → less thromboxane A2 → platelets can't clump for their lifespan (7–10 days).

INDICATIONS

- Prevention of MI (myocardial infarction) and stroke in angina/coronary artery disease
- Suspected acute MI — chew non-enteric aspirin (162–325 mg)

MAIN SIDE EFFECTS

- GI (gastrointestinal) upset, heartburn
- Easy bruising

ADVERSE EFFECTS

- GI bleeding/ulceration — report black tarry stools, coffee-ground vomit
- Salicylism (tinnitus) at high doses
- Reye's syndrome in children/teens with viral illness

CONTRAINDICATIONS & PRECAUTIONS

- Bleeding disorders, active bleeding
- Children/teens with viral illness (flu, chickenpox)
- Caution: peptic ulcer disease

INTERACTIONS

- Warfarin, other anticoagulants/antiplatelets → bleeding risk
- Ibuprofen → may block aspirin's cardioprotective effect
- Alcohol → GI bleeding risk

SPECIAL WARNINGS

- Stop about 7 days before elective surgery, as directed by provider
- Monitor for bleeding

CLIENT EDUCATION

- Take with food or water to limit stomach upset
- Report black stools, blood in vomit, unusual bruising
- Report ringing in the ears
- Don't add over-the-counter NSAIDs (nonsteroidal anti-inflammatory drugs) without asking

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Low-dose aspirin protects the heart; chew it (not enteric-coated) for a suspected MI.

Ch 22 · Cardiac Rhythm + Cholesterol

ANTIDYSRHYTHMICS

Procainamide

Ch 22 · Antidysrhythmics

Class IA antidysrhythmic (sodium channel blocker) · also: quinidine, disopyramide

How it works: Blocks sodium channels → slows conduction and prolongs repolarization in atria and ventricles.

INDICATIONS
- Atrial and ventricular dysrhythmias
- Life-threatening ventricular tachycardia

MAIN SIDE EFFECTS
- Nausea, vomiting, diarrhea
- Hypotension (especially IV)

ADVERSE EFFECTS
- Lupus-like syndrome (SLE, systemic lupus erythematosus) — joint pain, fever, rash; positive ANA (antinuclear antibody); report
- Blood dyscrasias (agranulocytosis, thrombocytopenia) — monitor CBC (complete blood count); report fever, sore throat, bleeding
- QRS widening and QT prolongation → new dysrhythmias (torsades de pointes)

CONTRAINDICATIONS & PRECAUTIONS
- Complete heart block
- Systemic lupus erythematosus
- Torsades de pointes
- Caution: heart failure, renal or hepatic impairment

INTERACTIONS
- Other QT-prolonging drugs → additive QT prolongation, torsades; monitor ECG (electrocardiogram)

SPECIAL WARNINGS
- **BLACK BOX:** positive ANA / lupus-like syndrome; blood dyscrasias
- Continuous ECG monitoring with IV — report QRS or QT widening
- Monitor blood pressure (hypotension with IV)
- Monitor CBC regularly

CLIENT EDUCATION
- Report joint pain, fever, or rash (lupus-like symptoms)
- Report sore throat, fever, unusual bruising or bleeding

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Procainamide = "pro-lupus": lupus-like syndrome and blood dyscrasias are the classic tested adverse effects.

Lidocaine (IV) *(Xylocaine)*

Ch 22 · Antidysrhythmics

Class IB antidysrhythmic (sodium channel blocker) · also: mexiletine

How it works: Blocks sodium channels in ventricles → slows conduction, decreases ventricular automaticity.

INDICATIONS
- Ventricular dysrhythmias (ventricular tachycardia, ventricular fibrillation)

MAIN SIDE EFFECTS
- Drowsiness
- Mild confusion
- Numbness/paresthesias

ADVERSE EFFECTS
- CNS (central nervous system) toxicity — drowsiness, confusion, paresthesias, tinnitus → seizures; stop infusion, notify provider
- Bradycardia, heart block, hypotension

CONTRAINDICATIONS & PRECAUTIONS
- Severe heart block (second- or third-degree) without pacemaker
- Caution: hepatic impairment, heart failure (slower clearance)

INTERACTIONS
- Cimetidine, beta blockers → increased lidocaine levels/toxicity; monitor for CNS effects

SPECIAL WARNINGS
- Given IV only for dysrhythmias (poor oral bioavailability — first-pass liver metabolism)
- Continuous ECG (electrocardiogram) monitoring; use infusion pump
- Assess neuro status often — early CNS changes signal toxicity
- Seizure precautions

CLIENT EDUCATION
- Tell the nurse right away about ringing in ears, numbness, confusion, or feeling drowsy

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Lidocaine toxicity = CNS first: drowsy, confused, tinnitus, tingling → seizures.

Flecainide

Ch 22 · Antidysrhythmics

Class IC antidysrhythmic (sodium channel blocker) · also: propafenone

How it works: Strongly blocks sodium channels → markedly slows conduction through the heart.

INDICATIONS
- Paroxysmal atrial fibrillation/flutter and PSVT (paroxysmal supraventricular tachycardia) without structural heart disease
- Life-threatening ventricular dysrhythmias

MAIN SIDE EFFECTS
- Dizziness
- Blurred vision
- Nausea

ADVERSE EFFECTS
- Proarrhythmic — can cause new or worse dysrhythmias
- Heart failure (negative inotrope)

CONTRAINDICATIONS & PRECAUTIONS
- Recent MI (myocardial infarction) or structural heart disease
- Cardiogenic shock
- Second- or third-degree heart block without pacemaker

SPECIAL WARNINGS
- **BLACK BOX:** proarrhythmic effects; increased mortality in post-MI clients (CAST trial)
- Start under ECG (electrocardiogram) monitoring

CLIENT EDUCATION
- Report palpitations, fainting, shortness of breath, or swelling

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Class IC = strongest sodium block; never in post-MI or structural heart disease.

Propranolol / esmolol *(Inderal / Brevibloc)*

Ch 22 · Antidysrhythmics

Class II antidysrhythmic (beta-adrenergic blocker) · also: acebutolol, metoprolol

How it works: Blocks beta-1 receptors in the heart → slows heart rate, slows AV (atrioventricular) conduction, decreases automaticity.

INDICATIONS
- Rate control in atrial fibrillation/flutter
- SVT (supraventricular tachycardia)
- Esmolol: short-acting IV for rapid, titratable rate control

MAIN SIDE EFFECTS
- Fatigue
- Dizziness

ADVERSE EFFECTS
- Bradycardia, AV block — hold and notify for low heart rate
- Hypotension
- Bronchospasm (propranolol is nonselective; esmolol is cardioselective)
- Heart failure worsening

CONTRAINDICATIONS & PRECAUTIONS
- Sinus bradycardia, second- or third-degree AV block
- Cardiogenic shock
- Propranolol: asthma/COPD (chronic obstructive pulmonary disease)
- Caution: diabetes (masks hypoglycemia)

INTERACTIONS
- Verapamil/diltiazem → additive bradycardia, heart block, hypotension; monitor ECG (electrocardiogram)

SPECIAL WARNINGS
- Check apical pulse and BP (blood pressure) before each dose
- Do not stop abruptly (rebound tachycardia, dysrhythmias)

CLIENT EDUCATION
- Do not stop suddenly
- Check pulse daily; report slow pulse or dizziness
- If diabetic: tachycardia warning sign of low glucose may be masked

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Esmolol = IV, very short half-life — turn it off and effects fade within minutes.

Amiodarone *(Pacerone, Nexterone)*

Ch 22 · Antidysrhythmics

Class III antidysrhythmic (potassium channel blocker) · also: dronedarone, sotalol, dofetilide, ibutilide

How it works: Blocks potassium channels → prolongs repolarization and refractory period; also slows heart rate and AV (atrioventricular) conduction.

INDICATIONS

- Life-threatening ventricular dysrhythmias (recurrent ventricular fibrillation, unstable ventricular tachycardia)
- Atrial fibrillation (off-label)

MAIN SIDE EFFECTS

- Nausea, anorexia, constipation
- Photosensitivity; blue-gray skin discoloration
- Corneal microdeposits — halos, photophobia, blurred vision
- Bradycardia, hypotension

ADVERSE EFFECTS

- Pulmonary toxicity — cough, dyspnea, crackles; baseline chest x-ray and PFTs (pulmonary function tests)
- Hepatotoxicity — monitor LFTs (liver function tests)
- Worsened dysrhythmias, heart block, QT prolongation
- Hypothyroidism or hyperthyroidism — monitor thyroid function
- Optic neuropathy — vision loss; regular eye exams

CONTRAINDICATIONS & PRECAUTIONS

- Sinus bradycardia, second- or third-degree heart block without pacemaker
- Cardiogenic shock
- Pregnancy, breastfeeding
- Caution: thyroid disease, liver disease, lung disease

INTERACTIONS

- Digoxin → increased digoxin levels/toxicity; reduce digoxin dose, monitor level
- Warfarin → increased INR (international normalized ratio)/bleeding; reduce warfarin dose, monitor INR
- Grapefruit juice → increased amiodarone levels/toxicity; avoid
- Other QT-prolonging drugs → torsades de pointes
- Beta blockers, verapamil, diltiazem → bradycardia, AV block, hypotension

SPECIAL WARNINGS

- **BLACK BOX:** pulmonary toxicity, hepatotoxicity, worsened dysrhythmias
- Very long half-life (weeks to months) — effects and toxicity persist after stopping
- Continuous ECG (electrocardiogram) monitoring during IV loading
- Baseline + periodic chest x-ray, PFTs, LFTs, thyroid tests, eye exams

CLIENT EDUCATION

- Report cough, shortness of breath, or chest pain right away
- Use sunscreen and protective clothing; skin may turn blue-gray
- Avoid grapefruit juice
- Report vision changes, yellow skin/eyes, dark urine
- Get regular eye exams and lab tests

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Amiodarone hits lungs, liver, thyroid, eyes, skin — and raises digoxin and warfarin levels.

Sotalol *(Betapace)*

Ch 22 · Antidysrhythmics

Class III antidysrhythmic (potassium channel blocker) with beta-blocking (Class II) activity

How it works: Blocks potassium channels → prolongs repolarization; also nonselective beta blocker → slows heart rate.

INDICATIONS

- Life-threatening ventricular dysrhythmias
- Maintenance of normal sinus rhythm in atrial fibrillation/flutter

MAIN SIDE EFFECTS

- Fatigue, dizziness
- Bradycardia

ADVERSE EFFECTS

- QT prolongation → torsades de pointes; monitor ECG (electrocardiogram)
- Bradycardia, AV (atrioventricular) block, hypotension
- Bronchospasm (nonselective beta blockade)

CONTRAINDICATIONS & PRECAUTIONS

- Baseline QT prolongation
- Asthma
- Sinus bradycardia, second- or third-degree heart block without pacemaker
- Severe renal impairment; Caution: renal impairment (dose by creatinine clearance)

INTERACTIONS

- Other QT-prolonging drugs → torsades
- Calcium channel blockers → additive bradycardia/hypotension

SPECIAL WARNINGS

- **BLACK BOX:** start (or restart) in a facility with continuous ECG monitoring for at least 3 days; calculate creatinine clearance
- Keep potassium and magnesium normal (low levels raise torsades risk)

CLIENT EDUCATION

- Report fainting, palpitations, or dizziness
- Do not stop suddenly

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Sotalol = beta blocker + Class III — start in the hospital on a monitor because of torsades.

Dofetilide *(Tikosyn)*

Ch 22 · Antidysrhythmics

Class III antidysrhythmic (potassium channel blocker)

How it works: Blocks potassium channels → prolongs repolarization and refractory period.

INDICATIONS

- Convert atrial fibrillation/flutter and maintain normal sinus rhythm

MAIN SIDE EFFECTS

- Headache
- Dizziness

ADVERSE EFFECTS

- QT prolongation → torsades de pointes

CONTRAINDICATIONS & PRECAUTIONS

- Baseline QT prolongation
- Severe renal impairment
- Hydrochlorothiazide, verapamil, cimetidine, ketoconazole, trimethoprim (raise dofetilide levels)

INTERACTIONS

- Drugs that raise dofetilide levels (verapamil, cimetidine, ketoconazole) → torsades; contraindicated together

SPECIAL WARNINGS

- **BLACK BOX:** start in a facility with continuous ECG (electrocardiogram) monitoring for at least 3 days; dose by creatinine clearance

CLIENT EDUCATION

- Report fainting or fast/irregular heartbeat
- Check with provider before starting any new medicine

ANTIDOTE

None specific: stop the drug and treat symptoms

Verapamil / diltiazem *(Calan / Cardizem)*

Ch 22 · Antidysrhythmics

Class IV antidysrhythmic (calcium channel blocker)

How it works: Blocks calcium channels in SA (sinoatrial) and AV (atrioventricular) nodes → slows heart rate and AV conduction.

INDICATIONS
- Rate control in atrial fibrillation/flutter
- SVT (supraventricular tachycardia)

MAIN SIDE EFFECTS
- Constipation (especially verapamil)
- Dizziness, headache
- Peripheral edema

ADVERSE EFFECTS
- Bradycardia, AV block
- Hypotension
- Heart failure worsening

CONTRAINDICATIONS & PRECAUTIONS
- Heart failure (reduced ejection fraction)
- Second- or third-degree AV block, sick sinus syndrome without pacemaker
- Severe hypotension

INTERACTIONS
- Beta blockers → additive bradycardia, AV block, hypotension; monitor
- Digoxin → verapamil raises digoxin levels; monitor level
- Grapefruit juice → increased drug levels; avoid

SPECIAL WARNINGS
- Check heart rate and BP (blood pressure) before dosing
- IV: continuous ECG (electrocardiogram) and BP monitoring

CLIENT EDUCATION
- Increase fiber and fluids to prevent constipation
- Report dizziness, slow pulse, swelling, or shortness of breath
- Avoid grapefruit juice

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Only the non-dihydropyridine CCBs (verapamil, diltiazem) slow the heart — nifedipine does not.

Adenosine *(Adenocard)*

Ch 22 · Antidysrhythmics

Other antidysrhythmic

How it works: Slows conduction through the AV (atrioventricular) node → interrupts reentry; half-life under 10 seconds.

INDICATIONS
- Convert PSVT (paroxysmal supraventricular tachycardia) to sinus rhythm

MAIN SIDE EFFECTS
- Facial flushing
- Shortness of breath, chest pressure
- Headache, dizziness

ADVERSE EFFECTS
- Brief asystole/pause for a few seconds — expected
- Bronchospasm
- Transient bradycardia, hypotension

CONTRAINDICATIONS & PRECAUTIONS
- Second- or third-degree AV block or sick sinus syndrome without pacemaker
- Caution: asthma (bronchospasm)

INTERACTIONS
- Methylxanthines (theophylline, caffeine) → block adenosine, reduce effect; may need higher dose
- Dipyridamole → increases adenosine effect; lower dose
- Carbamazepine → increased risk of heart block

SPECIAL WARNINGS
- Give rapid IV push over 1–2 seconds, then rapid normal saline flush
- Use port closest to the heart
- Continuous ECG (electrocardiogram) monitoring; crash cart available

CLIENT EDUCATION
- You may briefly feel flushing, chest pressure, or a pause — it passes in seconds

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Rapid push + flush, then expect a brief flat line — that's the drug working.

Digoxin *(Lanoxin)*

Ch 22 · Antidysrhythmics

Cardiac glycoside (used as antidysrhythmic)

How it works: Slows conduction through the AV (atrioventricular) node → slows ventricular rate; also increases contraction strength.

INDICATIONS
- Ventricular rate control in atrial fibrillation/flutter
- Heart failure

MAIN SIDE EFFECTS
- Nausea, anorexia

ADVERSE EFFECTS
- Digoxin toxicity — anorexia, nausea/vomiting, blurred vision, yellow-green halos, dysrhythmias
- Bradycardia — hold if apical pulse below 60 in adults and notify provider

CONTRAINDICATIONS & PRECAUTIONS
- Ventricular fibrillation
- Caution: hypokalemia, renal impairment, older adults

INTERACTIONS
- Potassium-wasting diuretics → hypokalemia → digoxin toxicity; monitor potassium
- Amiodarone, verapamil → increased digoxin levels; monitor level

SPECIAL WARNINGS
- Therapeutic level 0.5–2 ng/mL
- Take apical pulse for 1 full minute before giving
- Monitor potassium — hypokalemia increases toxicity

CLIENT EDUCATION
- Check pulse before each dose; report slow pulse
- Report loss of appetite, nausea, or vision changes

ANTIDOTE
- Digoxin immune fab (Digifab)

Remember: Low potassium = digoxin toxicity risk. Full card in the heart failure chapter.

Atropine

Ch 22 · Antidysrhythmics

Anticholinergic (muscarinic antagonist)

How it works: Blocks vagal (parasympathetic) effects on SA (sinoatrial) and AV (atrioventricular) nodes → increases heart rate.

INDICATIONS
- Symptomatic bradycardia

MAIN SIDE EFFECTS
- Dry mouth
- Blurred vision, photophobia
- Urinary retention
- Constipation

ADVERSE EFFECTS
- Tachycardia
- Anticholinergic toxicity — hot, dry, flushed skin, confusion, delirium

CONTRAINDICATIONS & PRECAUTIONS
- Angle-closure glaucoma
- Caution: BPH (benign prostatic hyperplasia)/urinary retention, myocardial ischemia

INTERACTIONS
- Other anticholinergic drugs (antihistamines, tricyclic antidepressants) → additive anticholinergic effects

SPECIAL WARNINGS
- Continuous ECG (electrocardiogram) and vital sign monitoring
- Monitor urine output

CLIENT EDUCATION
- Report trouble urinating, fast heartbeat, or eye pain

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Atropine speeds the heart by blocking the vagus nerve.

CHOLESTEROL (ANTILIPEMICS)

Atorvastatin *(Lipitor)*

HMG-CoA reductase inhibitor (statin) · also: simvastatin, lovastatin, pravastatin, rosuvastatin

How it works: Blocks HMG-CoA reductase → liver makes less cholesterol and removes more LDL (low-density lipoprotein) from blood.

INDICATIONS
- Lower LDL and total cholesterol; modestly raise HDL (high-density lipoprotein)
- Prevent MI (myocardial infarction), stroke in at-risk clients

MAIN SIDE EFFECTS
- Headache
- GI upset — abdominal pain, constipation, diarrhea
- Rash

ADVERSE EFFECTS
- Myopathy/rhabdomyolysis — muscle pain, tenderness, weakness; check CK (creatine kinase), stop drug if elevated
- Hepatotoxicity — baseline and periodic LFTs (liver function tests)

CONTRAINDICATIONS & PRECAUTIONS
- Pregnancy (stop before conceiving) and breastfeeding
- Active liver disease
- Caution: alcohol use disorder, renal impairment

INTERACTIONS
- Gemfibrozil and other fibrates → increased myopathy/rhabdomyolysis risk; monitor CK, report muscle pain
- Grapefruit juice → increased statin levels; avoid large amounts
- Macrolides, azole antifungals, cyclosporine, HIV protease inhibitors → increased statin levels/myopathy
- Some statins + warfarin → increased bleeding; monitor INR (international normalized ratio)

SPECIAL WARNINGS
- Monitor LFTs and lipid levels
- Monitor CK if muscle symptoms occur

CLIENT EDUCATION
- Report unexplained muscle pain, tenderness, or weakness right away
- Take simvastatin and lovastatin in the evening (lovastatin with evening meal); atorvastatin any time
- Avoid grapefruit juice
- Use reliable contraception; stop drug if pregnant
- Continue low-fat diet and exercise

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Muscle pain on a statin = check CK; statin + gemfibrozil = rhabdo risk.

Ezetimibe *(Zetia)*

Cholesterol absorption inhibitor

How it works: Blocks cholesterol absorption in the small intestine → less cholesterol reaches the liver → lower LDL (low-density lipoprotein).

INDICATIONS
- Lower LDL — alone or added to a statin

MAIN SIDE EFFECTS
- Diarrhea
- Joint pain, muscle aches
- Upper respiratory infection symptoms

ADVERSE EFFECTS
- Hepatotoxicity when combined with a statin — monitor LFTs (liver function tests)
- Myopathy (rare) — report muscle pain

CONTRAINDICATIONS & PRECAUTIONS
- With a statin: active liver disease, pregnancy, breastfeeding

INTERACTIONS
- Bile acid sequestrants → reduced ezetimibe absorption; take ezetimibe 2 h before or 4 h after
- Cyclosporine → increased ezetimibe levels; monitor

SPECIAL WARNINGS
- Monitor lipid levels; LFTs if combined with a statin

CLIENT EDUCATION
- Take once daily, with or without food
- Report muscle pain or weakness
- Continue low-fat diet and exercise

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Ezetimibe works in the gut (absorption); statins work in the liver (synthesis) — that's why they pair well.

Cholestyramine *(Questran)*

Bile acid sequestrant · also: colesevelam, colestipol

How it works: Binds bile acids in the intestine → excreted in stool → liver uses cholesterol to make new bile acids → lower LDL (low-density lipoprotein).

INDICATIONS
- Lower LDL (often with a statin)
- Cholestyramine: itching from partial biliary obstruction

MAIN SIDE EFFECTS
- Constipation
- Bloating, flatulence, nausea

ADVERSE EFFECTS
- Fat-soluble vitamin (A, D, E, K) deficiency
- Severe constipation/bowel obstruction (rare)

CONTRAINDICATIONS & PRECAUTIONS
- Complete biliary obstruction
- Bowel obstruction
- Caution: chronic constipation, high triglycerides (can raise triglycerides)

INTERACTIONS
- Binds many oral drugs (warfarin, digoxin, thiazides, thyroid hormone) → reduced absorption; take other drugs 1 h before or 4 h after
- Fat-soluble vitamins → reduced absorption; may need supplements

SPECIAL WARNINGS
- Not absorbed systemically — main problems are GI and drug binding
- Monitor lipid levels and bowel function

CLIENT EDUCATION
- Mix cholestyramine powder with fluid (water, juice) — never take it dry
- Take other medicines 1 hour before or 4 hours after
- Increase fluids and fiber to prevent constipation
- Colesevelam tablets: take with a meal and a drink

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Sequestrants "sequester" other drugs too — 1 hour before or 4 hours after.

Niacin (nicotinic acid) *(Niaspan)*

Vitamin B3 (antilipemic)

How it works: Decreases liver production of VLDL (very-low-density lipoprotein) → lowers LDL (low-density lipoprotein) and triglycerides; raises HDL (high-density lipoprotein).

INDICATIONS
- Hyperlipidemia — high triglycerides, high LDL, low HDL

MAIN SIDE EFFECTS
- Flushing of face and neck, itching, warmth
- GI upset — nausea, abdominal pain

ADVERSE EFFECTS
- Hepatotoxicity (higher risk with sustained-release) — monitor LFTs (liver function tests)
- Hyperglycemia — monitor glucose in diabetes
- Hyperuricemia/gout — monitor uric acid
- Myopathy when combined with a statin

CONTRAINDICATIONS & PRECAUTIONS
- Liver disease
- Active peptic ulcer disease
- Arterial bleeding
- Caution: diabetes, gout

INTERACTIONS
- Statins → increased myopathy/rhabdomyolysis risk; report muscle pain

SPECIAL WARNINGS
- Monitor LFTs, blood glucose, uric acid

CLIENT EDUCATION
- Take aspirin (325 mg) 30 minutes before dose to reduce flushing
- Flushing usually decreases after 1–2 weeks
- Avoid alcohol and hot drinks around dose (worsen flushing)
- Report yellow skin/eyes or dark urine

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Niacin: flushing (aspirin 30 min before), liver, high glucose, gout.

Gemfibrozil *(Lopid)*

Fibric acid derivative (fibrate) · also: fenofibrate

How it works: Increases breakdown of VLDL (very-low-density lipoprotein) → mainly lowers triglycerides; raises HDL (high-density lipoprotein).

INDICATIONS
- High triglycerides

MAIN SIDE EFFECTS
- GI upset — dyspepsia, abdominal pain, diarrhea

ADVERSE EFFECTS
- Gallstones (cholelithiasis) — report right upper quadrant pain
- Myopathy/rhabdomyolysis, especially with statins — check CK (creatine kinase)
- Hepatotoxicity — monitor LFTs (liver function tests)

CONTRAINDICATIONS & PRECAUTIONS
- Gallbladder disease
- Severe liver or kidney impairment

INTERACTIONS
- Warfarin → increased anticoagulant effect/bleeding; monitor INR (international normalized ratio), may lower warfarin dose
- Statins → increased myopathy/rhabdomyolysis risk; monitor CK

SPECIAL WARNINGS
- Monitor lipids, LFTs, CBC (complete blood count), and INR if on warfarin

CLIENT EDUCATION
- Take gemfibrozil 30 minutes before breakfast and dinner
- Report muscle pain, abdominal pain (right side), or unusual bleeding

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Fibrates: gallstones, myopathy, and they boost warfarin.

Evolocumab *(Repatha)*

PCSK9 inhibitor (monoclonal antibody) · also: alirocumab

How it works: Blocks PCSK9 protein → liver keeps more LDL (low-density lipoprotein) receptors → more LDL cleared from blood.

INDICATIONS
- Lower LDL in familial hypercholesterolemia or atherosclerotic heart disease, added to diet ± statin

MAIN SIDE EFFECTS
- Injection-site reactions — redness, pain, bruising
- Nasopharyngitis, flu-like symptoms
- Back pain

ADVERSE EFFECTS
- Hypersensitivity reactions — rash, hives; stop and notify provider

CONTRAINDICATIONS & PRECAUTIONS
- History of serious hypersensitivity to the drug

SPECIAL WARNINGS
- Subcutaneous only (every 2 weeks or monthly)
- Monitor lipid levels

CLIENT EDUCATION
- Inject subcutaneously into abdomen, thigh, or upper arm; rotate sites
- Store in refrigerator; let warm to room temperature about 30 minutes before injecting
- Report rash, hives, or swelling

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: PCSK9 inhibitors are injectable (subQ) — teach injection technique and site reactions.

Ch 33 · Nonopioid Analgesics

NSAIDS (NONSTEROIDAL ANTI-INFLAMMATORY DRUGS)

Aspirin (acetylsalicylic acid)

Ch 33 · NSAIDs (nonsteroidal anti-inflammatory drugs)

(Bayer, Ecotrin)

Salicylate; first-generation nonselective NSAID (nonsteroidal anti-inflammatory drug)

How it works: Irreversibly blocks COX-1 and COX-2 (cyclooxygenase) → less prostaglandins; antiplatelet effect lasts life of platelet.

INDICATIONS

- Mild to moderate pain, fever, inflammation
- Low dose (81 mg): prevention of MI (myocardial infarction) and stroke

MAIN SIDE EFFECTS

- GI (gastrointestinal) upset, heartburn, nausea

ADVERSE EFFECTS

- GI ulcers and bleeding — report black tarry stools, coffee-ground vomit
- Salicylism: tinnitus, hearing loss, headache, dizziness, sweating
- Toxicity: hyperthermia, respiratory alkalosis then metabolic acidosis
- Bleeding (prolonged bleeding time)
- Renal impairment
- Reye's syndrome in children/teens

CONTRAINDICATIONS & PRECAUTIONS

- Children and teens with viral illness (flu, chickenpox) — Reye's syndrome
- Bleeding disorders (e.g., hemophilia), vitamin K deficiency
- PUD (peptic ulcer disease)
- Pregnancy (especially 3rd trimester)
- Caution: asthma (aspirin sensitivity), alcohol use, renal impairment

INTERACTIONS

- Warfarin/anticoagulants → increased bleeding
- Alcohol, glucocorticoids → GI bleeding/ulcers
- Ibuprofen → blocks aspirin's antiplatelet effect; take aspirin first
- Ibuprofen/other NSAIDs → more GI and bleeding risk

SPECIAL WARNINGS

- Stop 7 days (1 week) before elective surgery
- Toxicity treatment: no antidote — activated charcoal, cooling, IV sodium bicarbonate, hemodialysis
- Monitor for bleeding and tinnitus

CLIENT EDUCATION

- Take with food, milk, or full glass of water
- Do not give to children/teens with flu or chickenpox symptoms
- Report ringing in ears, black stools, unusual bruising
- Avoid alcohol

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Tinnitus = first sign of salicylism. Kids + virus + aspirin = Reye's.

Ibuprofen *(Advil, Motrin)*

Ch 33 · NSAIDs (nonsteroidal anti-inflammatory drugs)

First-generation nonselective NSAID (nonsteroidal anti-inflammatory drug) · also: Naproxen (Aleve, Naprosyn), indomethacin, ketorolac

How it works: Reversibly blocks COX-1 and COX-2 (cyclooxygenase) → less prostaglandins → less pain, fever, inflammation.

INDICATIONS

- Mild to moderate pain, fever
- Inflammation (arthritis)
- Dysmenorrhea

MAIN SIDE EFFECTS

- GI (gastrointestinal) upset, dyspepsia, nausea
- Dizziness, headache

ADVERSE EFFECTS

- GI ulcers, bleeding, perforation
- Acute kidney injury — monitor BUN (blood urea nitrogen), creatinine, I&O (intake and output)
- MI (myocardial infarction), stroke
- Fluid retention, hypertension, heart failure worsening

CONTRAINDICATIONS & PRECAUTIONS

- Perioperative pain with CABG (coronary artery bypass graft)
- Pregnancy at 20 weeks or later — avoid, especially 3rd trimester (premature ductus arteriosus closure)
- PUD (peptic ulcer disease), active GI bleed
- Caution: renal impairment, heart failure, hypertension, older adults
- Caution: bleeding disorders, alcohol use

INTERACTIONS

- Anticoagulants, aspirin → bleeding
- Glucocorticoids, alcohol → GI ulcers/bleeding
- Lithium → higher lithium levels
- ACE (angiotensin-converting enzyme) inhibitors, diuretics → less BP effect, kidney injury risk

SPECIAL WARNINGS

- **BLACK BOX:** increased risk of CV (cardiovascular) thrombotic events — MI, stroke
- **BLACK BOX:** GI bleeding, ulceration, perforation — can occur without warning
- Use lowest effective dose for shortest time
- Monitor renal function and for GI bleeding

CLIENT EDUCATION

- Take with food or milk
- Report black stools, coffee-ground vomit, chest pain, weakness on one side
- Report decreased urine or swelling
- Avoid alcohol; check OTC labels for duplicate NSAIDs

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Naproxen works longer than ibuprofen (twice daily). All nonselective NSAIDs: heart, gut, kidneys.

Ketorolac *(Toradol)*

Ch 33 · NSAIDs (nonsteroidal anti-inflammatory drugs)

First-generation nonselective NSAID (nonsteroidal anti-inflammatory drug)

How it works: Nonselective COX (cyclooxygenase) inhibitor with strong analgesic effect — opioid-level pain relief without respiratory depression.

INDICATIONS
- Short-term moderately severe acute pain (e.g., post-op)

MAIN SIDE EFFECTS
- GI (gastrointestinal) upset, nausea
- Headache, drowsiness

ADVERSE EFFECTS
- Renal impairment / acute kidney injury
- GI bleeding, ulcers
- Bleeding (inhibits platelets)
- CV (cardiovascular) events — MI, stroke

CONTRAINDICATIONS & PRECAUTIONS
- Renal impairment or risk of hypovolemia
- Active GI bleed, PUD (peptic ulcer disease)
- Bleeding risk; before major surgery
- Labor and delivery; CABG (coronary artery bypass graft) pain
- Use with aspirin or other NSAIDs

INTERACTIONS
- Aspirin/other NSAIDs → contraindicated together
- Anticoagulants → bleeding
- Probenecid, pentoxifylline → contraindicated

SPECIAL WARNINGS
- **BLACK BOX:** max 5 days total (IV/IM + oral combined)
- **BLACK BOX:** GI bleeding, CV events, renal risk, bleeding
- Routes: IV, IM, oral, intranasal; oral only as continuation of IV/IM
- Monitor BUN (blood urea nitrogen), creatinine, urine output

CLIENT EDUCATION
- Do not take for more than 5 days
- Report bleeding, black stools, decreased urination

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Ketorolac = 5-day limit, kidneys at risk.

Celecoxib *(Celebrex)*

Ch 33 · NSAIDs (nonsteroidal anti-inflammatory drugs)

Second-generation NSAID — COX-2 (cyclooxygenase-2) selective inhibitor

How it works: Blocks only COX-2 → reduces pain and inflammation with less GI (gastrointestinal) damage; no antiplatelet effect.

INDICATIONS
- Osteoarthritis, rheumatoid arthritis
- Acute pain, dysmenorrhea

MAIN SIDE EFFECTS
- Dyspepsia, abdominal pain
- Edema

ADVERSE EFFECTS
- MI (myocardial infarction), stroke
- GI bleeding (lower risk than nonselective, but still possible)
- Renal impairment
- Hypertension, fluid retention

CONTRAINDICATIONS & PRECAUTIONS
- Sulfonamide (sulfa) allergy
- Perioperative pain with CABG (coronary artery bypass graft)
- Pregnancy at 20 weeks or later / 3rd trimester
- Caution: CV disease, renal impairment, history of GI bleed

INTERACTIONS
- Warfarin → increased bleeding risk
- Lithium → higher lithium levels
- ACE (angiotensin-converting enzyme) inhibitors, diuretics → less BP effect

SPECIAL WARNINGS
- **BLACK BOX:** CV (cardiovascular) thrombotic events — MI, stroke
- **BLACK BOX:** GI bleeding, ulceration, perforation
- Does NOT protect the heart like aspirin (no antiplatelet effect)

CLIENT EDUCATION
- Tell provider about sulfa allergy
- Report chest pain, shortness of breath, black stools, swelling
- Lowest dose, shortest time

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Celecoxib: COX-2 only → easier on stomach, harder on heart; sulfa allergy = no.

ACETAMINOPHEN

Acetaminophen *(Tylenol)*

Ch 33 · Acetaminophen

Nonopioid analgesic / antipyretic (not an NSAID)

How it works: Inhibits prostaglandin synthesis mainly in the CNS (central nervous system) → reduces pain and fever; no anti-inflammatory effect.

INDICATIONS
- Mild to moderate pain
- Fever
- Preferred for children with viral illness (no Reye's risk)

MAIN SIDE EFFECTS
- Few at therapeutic doses

ADVERSE EFFECTS
- Hepatotoxicity / liver failure with overdose
- Early overdose signs: nausea, vomiting, sweating, abdominal pain
- Later: jaundice, liver failure
- Rare serious skin reactions (Stevens-Johnson syndrome) — stop if rash

CONTRAINDICATIONS & PRECAUTIONS
- Severe liver disease
- Caution: chronic alcohol use, malnutrition
- Caution: renal impairment

INTERACTIONS
- Alcohol → increased hepatotoxicity
- Warfarin → increased INR (international normalized ratio) with regular use
- Combination products (opioid + acetaminophen, cold/flu meds) → hidden extra doses

SPECIAL WARNINGS
- Max 4 g/day healthy adults; many sources advise ≤ 3 g/day
- Chronic alcohol use: lower limit (often ≤ 2 g/day)
- Monitor LFTs (liver function tests) with long-term use
- No anti-inflammatory or antiplatelet effect; no GI bleeding

CLIENT EDUCATION
- Read labels — acetaminophen is in many combination products
- Avoid alcohol
- Do not exceed daily maximum
- Use correct measuring device for children's liquid doses

ANTIDOTE
- Acetylcysteine

Remember: Overdose → acetylcysteine (oral or IV; rotten-egg smell — mix with juice).

OPIOID AGONISTS

Morphine *(MS Contin)*

Ch 34 · Opioid agonists

Opioid agonist (strong mu agonist), Schedule II · also: Fentanyl, hydromorphone, oxycodone, codeine, meperidine, methadone, hydrocodone

How it works: Binds mu (and kappa) opioid receptors in the CNS (central nervous system) → blocks pain transmission and perception.

INDICATIONS
- Moderate to severe acute or chronic pain
- Myocardial infarction pain (reduces cardiac workload)
- Pre-op sedation / adjunct to anesthesia
- Dyspnea in end-of-life care

MAIN SIDE EFFECTS
- Constipation (does not go away with tolerance)
- Nausea/vomiting
- Sedation, dizziness
- Miosis (pinpoint pupils)
- Itching (histamine release)

ADVERSE EFFECTS
- Respiratory depression — hold if respirations < 12/min; give naloxone if severe
- Orthostatic hypotension — rise slowly, fall precautions
- Urinary retention — monitor I&O (intake and output), bladder scan
- Biliary colic (spasm of biliary tract)
- Cough suppression — encourage cough, deep breathing post-op
- Toxicity triad: coma, respiratory depression, pinpoint pupils

CONTRAINDICATIONS & PRECAUTIONS
- After biliary tract surgery
- Severe asthma / emphysema / respiratory disorders
- Caution: head injury (raises ICP, intracranial pressure; masks neuro signs)
- Caution: hypotension, BPH (benign prostatic hyperplasia), liver or kidney impairment
- Caution: older adults, pregnancy, inflammatory bowel disease

INTERACTIONS
- CNS depressants (alcohol, benzodiazepines, barbiturates) → additive respiratory depression; avoid
- Anticholinergics (antihistamines, TCAs, tricyclic antidepressants) → worse constipation, urinary retention
- MAOIs (monoamine oxidase inhibitors) → hypertensive crisis, hyperpyrexia; avoid
- Antihypertensives → additive hypotension
- Agonist-antagonists (e.g., pentazocine) → can precipitate withdrawal

SPECIAL WARNINGS
- **BLACK BOX:** life-threatening respiratory depression
- **BLACK BOX:** addiction, abuse, misuse; neonatal opioid withdrawal syndrome with prolonged use in pregnancy
- **BLACK BOX:** risk with benzodiazepines/CNS depressants; accidental ingestion fatal
- Check respiratory rate, sedation level, BP (blood pressure), pain before and after doses
- PCA (patient-controlled analgesia): only the client presses the button; check pump settings with 2nd nurse
- Do not crush or chew extended-release tablets

CLIENT EDUCATION
- Get up slowly; call for help to walk
- Increase fluids and fiber; use stool softener + stimulant laxative
- Avoid alcohol and other sedating drugs
- Do not drive until you know how this affects you
- Report trouble breathing, extreme sleepiness, or trouble urinating

ANTIDOTE
- Naloxone

Remember: Respirations < 12/min → hold the dose. Constipation and miosis are the two effects that do NOT develop tolerance.

Fentanyl *(Duragesic (patch), Sublimaze (IV))*

Ch 34 · Opioid agonists

Opioid agonist (strong mu agonist), Schedule II

How it works: Highly potent mu agonist (about 50–100x morphine) → blocks pain perception; lipid-soluble so it absorbs through skin.

INDICATIONS
- Chronic severe pain in opioid-tolerant clients (transdermal patch)
- Anesthesia / procedural sedation (IV)
- Breakthrough cancer pain (buccal, sublingual)

MAIN SIDE EFFECTS
- Sedation, dizziness
- Nausea, constipation
- Skin irritation at patch site

ADVERSE EFFECTS
- Respiratory depression — can be fatal, especially in opioid-naive clients
- Muscle (chest wall) rigidity with rapid IV push
- Bradycardia, hypotension

CONTRAINDICATIONS & PRECAUTIONS
- Patch: opioid-naive clients
- Patch: acute, short-term, or post-op pain
- Caution: fever or external heat (increases absorption)
- Caution: respiratory disease, older adults

INTERACTIONS
- CNS depressants (alcohol, benzodiazepines) → respiratory depression
- CYP3A4 inhibitors (e.g., ketoconazole, ritonavir, grapefruit) → raise fentanyl levels, overdose risk

SPECIAL WARNINGS
- **BLACK BOX:** patch only for opioid-tolerant clients — respiratory depression and death
- **BLACK BOX:** heat exposure increases absorption → overdose
- **BLACK BOX:** accidental exposure (children, pets) can be fatal
- Patch changed every 72 h; takes 12–24 h to reach full effect — give short-acting analgesic meanwhile
- Drug remains in skin ~ 12+ h after patch removal — keep monitoring
- Remove old patch before applying new; rotate sites

CLIENT EDUCATION
- Apply to clean, dry, hairless skin on upper body; press 30 seconds
- Avoid heating pads, electric blankets, hot tubs, saunas; report fever
- Fold used patch sticky sides together, then dispose as instructed (flush per FDA list)
- Keep patches away from children and pets
- Do not cut the patch

ANTIDOTE
- Naloxone

Remember: Patch = 72 h, opioid-tolerant only, no heat, fold and dispose.

Hydromorphone *(Dilaudid)*

Ch 34 · Opioid agonists

Opioid agonist (strong mu agonist), Schedule II

How it works: Mu agonist, several times more potent than morphine → blocks pain perception.

INDICATIONS
- Moderate to severe pain
- Cough (rarely)

MAIN SIDE EFFECTS
- Sedation, dizziness
- Nausea/vomiting, constipation
- Itching

ADVERSE EFFECTS
- Respiratory depression — hold if respirations < 12/min
- Hypotension
- Same adverse effects as morphine

CONTRAINDICATIONS & PRECAUTIONS
- Severe respiratory depression, acute/severe asthma
- High-potency (HP) formulation: opioid-naive clients
- Caution: head injury, older adults, liver/kidney impairment

INTERACTIONS
- CNS depressants (alcohol, benzodiazepines) → respiratory depression
- Anticholinergics → constipation, urinary retention
- MAOIs (monoamine oxidase inhibitors) → avoid

SPECIAL WARNINGS
- HIGH-ALERT medication — independent double check
- **BLACK BOX:** dosing errors — confusion between mg and mL and between standard vs high-potency concentrations → overdose
- Never substitute mg-for-mg with morphine
- **BLACK BOX:** respiratory depression, addiction, neonatal withdrawal (same as other opioids)

CLIENT EDUCATION
- Same as morphine: rise slowly, prevent constipation, avoid alcohol
- Report slow or shallow breathing, extreme sleepiness

ANTIDOTE
- Naloxone

Remember: Hydromorphone ≠ morphine — much more potent. Mixing them up is a classic fatal error.

Oxycodone *(OxyContin (ER); Percocet (with acetaminophen))*

Ch 34 · Opioid agonists

Opioid agonist (strong mu agonist), Schedule II · also: Hydrocodone (often combined with acetaminophen)

How it works: Mu agonist → blocks pain perception; oral, immediate- or extended-release.

INDICATIONS
- Moderate to severe pain
- ER (extended-release): around-the-clock chronic pain

MAIN SIDE EFFECTS
- Constipation
- Nausea, sedation, dizziness
- Itching

ADVERSE EFFECTS
- Respiratory depression
- Orthostatic hypotension
- Overdose if ER tablet crushed, chewed, or broken

CONTRAINDICATIONS & PRECAUTIONS
- Severe respiratory depression, acute/severe asthma
- GI (gastrointestinal) obstruction / paralytic ileus
- ER form: not for as-needed or acute pain

INTERACTIONS
- CNS depressants (alcohol, benzodiazepines) → respiratory depression
- CYP3A4 inhibitors → higher oxycodone levels

SPECIAL WARNINGS
- **BLACK BOX:** respiratory depression, addiction/abuse, neonatal withdrawal
- Do NOT crush, chew, or cut ER tablets — releases whole dose at once → overdose, death
- Combination products contain acetaminophen — count toward daily acetaminophen max

CLIENT EDUCATION
- Swallow ER tablets whole
- Check other meds for hidden acetaminophen
- Prevent constipation; avoid alcohol

ANTIDOTE
- Naloxone

Remember: ER = swallow whole. Percocet = oxycodone + acetaminophen (watch liver).

Codeine

Ch 34 · Opioid agonists

Opioid agonist (moderate to strong), Schedule II alone; lower schedules in combinations

How it works: Prodrug — converted by liver enzyme CYP2D6 into morphine, which provides analgesia.

INDICATIONS
- Mild to moderate pain (often with acetaminophen)
- Cough suppression (antitussive)

MAIN SIDE EFFECTS
- Constipation
- Nausea, sedation, dizziness

ADVERSE EFFECTS
- Respiratory depression — especially in ultrarapid CYP2D6 metabolizers (too much morphine made fast)
- Death in children from respiratory depression

CONTRAINDICATIONS & PRECAUTIONS
- Children under 12 years
- Children under 18 after tonsillectomy and/or adenoidectomy
- Breastfeeding — infant can get toxic morphine levels
- Caution: respiratory disease

INTERACTIONS
- CNS depressants (alcohol, benzodiazepines) → respiratory depression
- CYP2D6 inhibitors (e.g., fluoxetine, paroxetine) → less analgesia

SPECIAL WARNINGS
- **BLACK BOX:** ultrarapid metabolism → life-threatening respiratory depression and death in children after tonsillectomy/adenoidectomy
- Breastfeeding infants: watch for sleepiness, poor feeding, limpness, breathing problems
- Also BLACK BOX: addiction, respiratory depression, neonatal withdrawal

CLIENT EDUCATION
- Do not use while breastfeeding
- Combination products may contain acetaminophen — count it
- Prevent constipation; avoid alcohol

ANTIDOTE
- Naloxone

Remember: Codeine works only after the liver turns it into morphine — fast converters (ultrarapid metabolizers) overdose.

Meperidine *(Demerol)*

Ch 34 · Opioid agonists

Opioid agonist (strong mu agonist), Schedule II

How it works: Mu agonist → blocks pain perception; metabolite normeperidine is toxic and accumulates.

INDICATIONS
- Short-term moderate to severe pain (not first-line; not for long-term use)

MAIN SIDE EFFECTS
- Sedation, dizziness
- Nausea, constipation
- Anticholinergic effects: dry mouth, tachycardia, urinary retention

ADVERSE EFFECTS
- Normeperidine toxicity → CNS excitation: anxiety, tremors, seizures
- Respiratory depression

CONTRAINDICATIONS & PRECAUTIONS
- Use of MAOIs (monoamine oxidase inhibitors) within 14 days
- Caution/avoid: older adults
- Caution/avoid: renal impairment (normeperidine builds up)
- Caution: seizure disorder

INTERACTIONS
- MAOIs → serotonin syndrome / hyperpyrexia, can be fatal; contraindicated
- CNS depressants → respiratory depression

SPECIAL WARNINGS
- Normeperidine has long half-life — accumulates with repeated doses, renal impairment
- Monitor for tremors, twitching, seizures
- Limit to short-term use (e.g., ≤ 48 h)

CLIENT EDUCATION
- Report shakiness, twitching, agitation
- Avoid alcohol; rise slowly

ANTIDOTE
- Naloxone (does not reverse normeperidine seizures)

Remember: Meperidine = seizures from normeperidine; avoid in older adults and kidney disease; never with MAOIs.

Methadone *(Dolophine)*

Ch 34 · Opioid agonists

Opioid agonist (strong mu agonist), Schedule II

How it works: Long-acting mu agonist → blocks pain and prevents opioid withdrawal/cravings.

INDICATIONS
- Opioid use disorder — detox and maintenance (certified opioid treatment programs)
- Severe chronic pain

MAIN SIDE EFFECTS
- Constipation
- Sedation, dizziness, nausea
- Sweating

ADVERSE EFFECTS
- QT prolongation → torsades de pointes (dysrhythmia)
- Respiratory depression — delayed, from drug accumulation

CONTRAINDICATIONS & PRECAUTIONS
- Severe respiratory depression, acute/severe asthma
- Caution: existing QT prolongation, low potassium/magnesium
- Caution: other QT-prolonging drugs

INTERACTIONS
- Other QT-prolonging drugs → torsades risk
- CNS depressants (alcohol, benzodiazepines) → respiratory depression
- CYP3A4 inhibitors raise levels; inducers (e.g., rifampin) can cause withdrawal

SPECIAL WARNINGS
- **BLACK BOX:** QT prolongation — baseline and periodic ECG (electrocardiogram)
- **BLACK BOX:** respiratory depression — peak effect lags behind pain relief; long half-life → accumulation
- Titrate slowly; monitor closely when starting or changing dose
- Also BLACK BOX: addiction, neonatal withdrawal, accidental ingestion

CLIENT EDUCATION
- Take exactly as prescribed — do not take extra for pain
- Report palpitations, fainting, dizziness
- Avoid alcohol and sedatives

ANTIDOTE
- Naloxone (may need repeated doses — methadone outlasts naloxone)

Remember: Methadone: long half-life + QT prolongation. Pain relief wears off before the respiratory risk does.

Tramadol *(Ultram)*

Ch 34 · Opioid agonists

Opioid agonist (weak mu agonist), Schedule IV

How it works: Weak mu agonist plus blocks reuptake of serotonin and norepinephrine → pain relief.

INDICATIONS
- Moderate to moderately severe pain

MAIN SIDE EFFECTS
- Dizziness, sedation, headache
- Nausea, constipation, dry mouth

ADVERSE EFFECTS
- Seizures — risk higher with high doses or seizure history
- Serotonin syndrome (agitation, hyperthermia, tremor, tachycardia)
- Respiratory depression (less than morphine)

CONTRAINDICATIONS & PRECAUTIONS
- Children under 12; under 18 after tonsillectomy/adenoidectomy
- Caution: seizure disorder
- Caution: other serotonergic drugs
- Caution: suicide risk, alcohol/opioid misuse

INTERACTIONS
- SSRIs (selective serotonin reuptake inhibitors), SNRIs, TCAs, MAOIs, triptans → serotonin syndrome
- Drugs that lower seizure threshold → seizures
- CNS depressants → respiratory depression

SPECIAL WARNINGS
- **BLACK BOX:** ultrarapid CYP2D6 metabolism → respiratory depression in children (same as codeine)
- **BLACK BOX:** addiction, respiratory depression, neonatal withdrawal
- Can still cause dependence and withdrawal

CLIENT EDUCATION
- Report shaking, agitation, fever, fast heartbeat
- Avoid alcohol
- Do not exceed prescribed dose

ANTIDOTE
- Naloxone (partial reversal; may increase seizure risk)

Remember: Tramadol = weak opioid + serotonin effects → watch for seizures and serotonin syndrome.

Opioids — general nursing care & client education

Ch 34 · Opioid agonists

Applies to all opioid agonists

How it works: All opioid agonists share sedation, respiratory depression, constipation, and dependence risks.

INDICATIONS
- Pain management across acute, chronic, and end-of-life settings

MAIN SIDE EFFECTS
- Constipation, nausea, sedation, itching, urinary retention
- Orthostatic hypotension → fall risk

ADVERSE EFFECTS
- Sedation increases BEFORE respiratory depression — use a sedation scale (e.g., POSS, Pasero Opioid-Induced Sedation Scale)
- Withdrawal (abstinence syndrome): yawning, runny nose, sweating, gooseflesh, dilated pupils
- Withdrawal also: abdominal cramps, diarrhea, nausea, irritability, muscle aches, restlessness

CONTRAINDICATIONS & PRECAUTIONS
- Caution: older adults, respiratory disease, head injury, liver/kidney impairment
- Caution: history of substance use disorder

INTERACTIONS
- Alcohol, benzodiazepines, other CNS depressants → respiratory depression

SPECIAL WARNINGS
- Assess pain, respiratory rate, sedation level before and after each dose
- Tolerance = need higher dose for same effect (not addiction)
- Physical dependence = withdrawal if stopped abruptly — taper off
- Addiction = compulsive use despite harm (psychological)
- Start constipation prevention with first dose: stool softener + stimulant laxative (e.g., docusate + senna)
- Fall precautions: bed low, call light in reach, assist with ambulation

CLIENT EDUCATION
- No alcohol or other sedating drugs
- Change positions slowly; ask for help walking
- Drink fluids, eat fiber, take your bowel regimen
- Do not stop suddenly after long-term use — taper with your provider
- Store securely; dispose of unused opioids safely

ANTIDOTE
- Naloxone

Remember: Watch sedation first — a client getting sleepier is the early warning before breathing slows.

OPIOID ANTAGONISTS

Naloxone *(Narcan)*

Ch 34 · Opioid antagonists

Opioid antagonist (pure)

How it works: Competitively blocks mu opioid receptors → reverses opioid effects (respiratory depression, sedation, analgesia).

INDICATIONS
- Opioid overdose / respiratory depression
- Reversal of post-op opioid sedation
- Neonatal opioid-induced respiratory depression

MAIN SIDE EFFECTS
- Few effects if no opioids present

ADVERSE EFFECTS
- Abrupt withdrawal in opioid-dependent clients
- Return of severe pain (reverses analgesia)
- Tachycardia, hypertension, dysrhythmias (rare), nausea/vomiting

CONTRAINDICATIONS & PRECAUTIONS
- Caution: opioid dependence (precipitates withdrawal)
- Caution: cardiovascular disease

INTERACTIONS
- Opioid agonists → effects reversed (intended action)

SPECIAL WARNINGS
- Routes: IV (fastest, 1–2 min), IM, subcutaneous, intranasal (OTC, over-the-counter)
- Short duration — often shorter than the opioid → re-sedation; monitor closely and redose
- Repeat doses every 2–3 min as needed
- Chronic opioid users: dilute and titrate slowly to restore breathing but keep pain control
- Have resuscitation equipment ready; monitor respiratory rate, SpO2 (oxygen saturation), LOC (level of consciousness)

CLIENT EDUCATION
- Family of high-risk clients: keep intranasal naloxone available, call 911 after giving it
- Effects may wear off before the opioid does — stay with the person

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Naloxone wears off fast → client can slip back into respiratory depression. Monitor and redose.

Naltrexone *(Vivitrol (IM extended-release))*

Ch 34 · Opioid antagonists

Opioid antagonist (pure, long-acting)

How it works: Blocks opioid receptors for a long time → no euphoria from opioids; reduces alcohol cravings.

INDICATIONS
- Alcohol use disorder
- Opioid use disorder (prevents relapse after detox)

MAIN SIDE EFFECTS
- Nausea, headache, dizziness, fatigue
- Injection site reactions (IM)

ADVERSE EFFECTS
- Hepatotoxicity — monitor LFTs (liver function tests)
- Precipitated withdrawal if opioids still in system
- Overdose if client tries to override the blockade or relapses after stopping (lost tolerance)

CONTRAINDICATIONS & PRECAUTIONS
- Current opioid use or opioid dependence
- Acute opioid withdrawal
- Acute hepatitis or liver failure
- Client who needs opioids for pain

INTERACTIONS
- Opioid analgesics, opioid cough/antidiarrheal meds → blocked (no effect)

SPECIAL WARNINGS
- Must be opioid-free 7–10 days before starting
- Routes: oral daily or IM extended-release once a month (gluteal)
- Pain management: opioids will not work — need non-opioid options
- Monitor liver function

CLIENT EDUCATION
- Carry ID/wallet card stating you take naltrexone
- Taking large opioid doses to overcome the block can be fatal
- After stopping, tolerance is lower — usual old dose can cause overdose
- Report yellow skin/eyes, dark urine, abdominal pain

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Naltrexone = maintenance, naloxone = emergency. Opioid-free 7–10 days first.

Methylnaltrexone *(Relistor)*

Ch 34 · Opioid antagonists

Peripherally acting mu opioid receptor antagonist

How it works: Blocks opioid receptors in the gut only (does not cross blood-brain barrier) → relieves constipation without reversing analgesia.

INDICATIONS
- Opioid-induced constipation (chronic non-cancer pain or advanced illness)

MAIN SIDE EFFECTS
- Abdominal pain, cramping
- Flatulence, nausea, diarrhea
- Dizziness, sweating

ADVERSE EFFECTS
- GI (gastrointestinal) perforation — report severe or persistent abdominal pain
- Severe diarrhea — stop and notify provider

CONTRAINDICATIONS & PRECAUTIONS
- Known or suspected mechanical GI obstruction
- Caution: GI lesions (ulcers, diverticulitis, cancer)
- Caution: severe renal impairment (dose adjust)

INTERACTIONS
- Other opioid antagonists → additive effects, possible withdrawal

SPECIAL WARNINGS
- Does not reverse pain relief or cause central withdrawal
- Routes: subcutaneous injection or oral
- Often works within hours (subcut can be within 30 min) — be near a bathroom

CLIENT EDUCATION
- Stay close to a toilet after a dose
- Stop and report severe diarrhea or severe belly pain
- Stop other laxatives only if told to

ANTIDOTE

None specific: stop the drug and treat symptoms

Remember: Methylnaltrexone = gut-only blocker → treats opioid constipation, pain control stays.